

Clustering

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Unsupervised learning

- What is clustering? 
 - K-means clustering 
 - DBSCAN clustering
 - Hierarchical clustering
 - Evaluation methods for clustering 
- 

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Unsupervised Algo $\Rightarrow$ only features are present
there are no labels

We focus on finding the patterns in data

F ₁	f ₂	F ₃	f ₄	(Y)X

What is Clustering?

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C1 and C2

Unsupervised Learning

- Un-supervised learning is the type of ML algorithms in which ML model identifies patterns in the data
- It automatically identifies group of similar examples

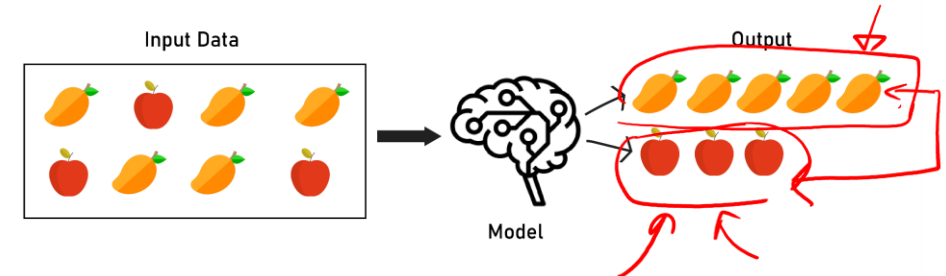
E.g.,

Clustering models

Principal Component Analysis (PCA)

cluster is a group of similar data points

how do we measure the similarity between the data points??
UNSUPERVISED LEARNING IN ML



Color	Weight	Dimensions	Smell

In a cluster, all the data points are similar to each other.

But these data points are dissimilar to all other data points not lying in the cluster

What is clustering?

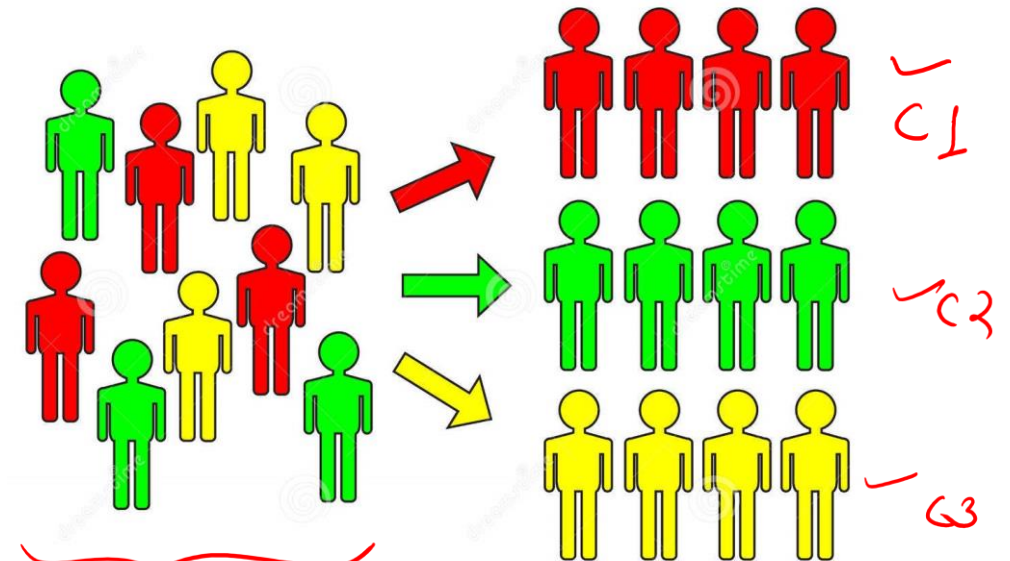
Give tailored products to the consumer

Goal 1 Similar data points should be in the same group

Goal 2 Dis-similar data points should be in different groups

Customer Segmentation

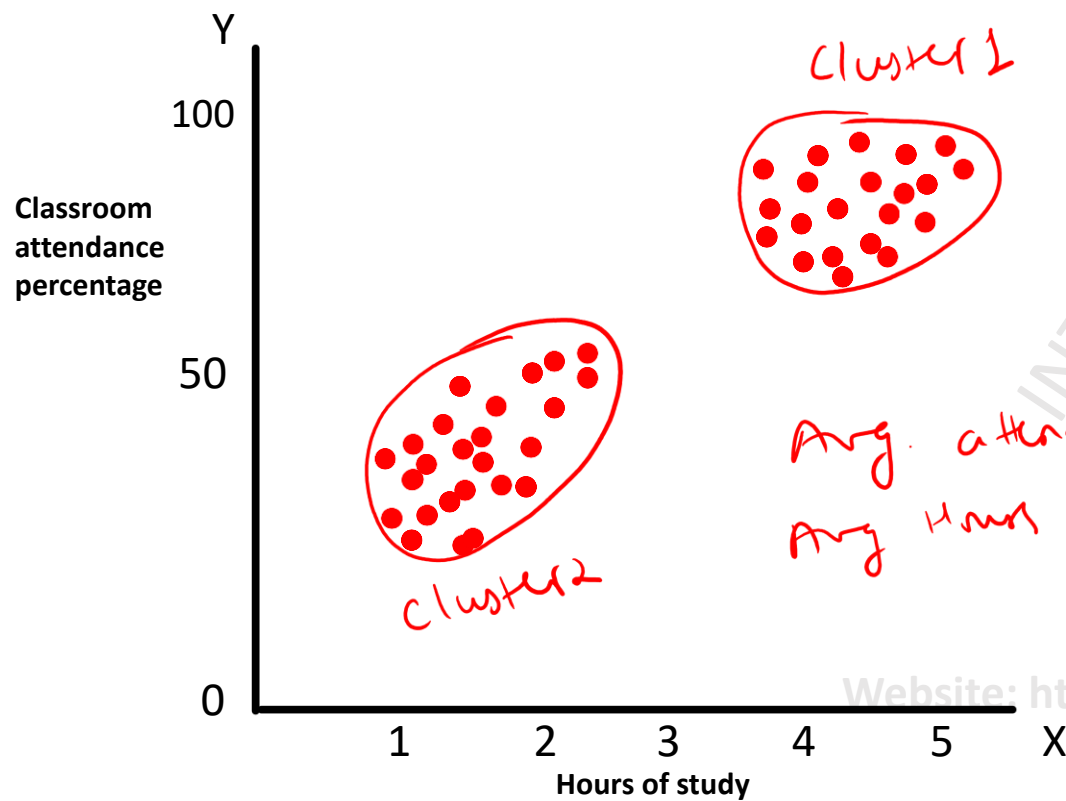
- Clustering is grouping of data points in such a way that similar data points are in same group and dis-similar data points are in different groups
- It is unsupervised learning method – labels are not available to train the model
- E.g., use cases:
 - Customer segmentation – identify customers with similar interests
 - Identify themes in customer complaints



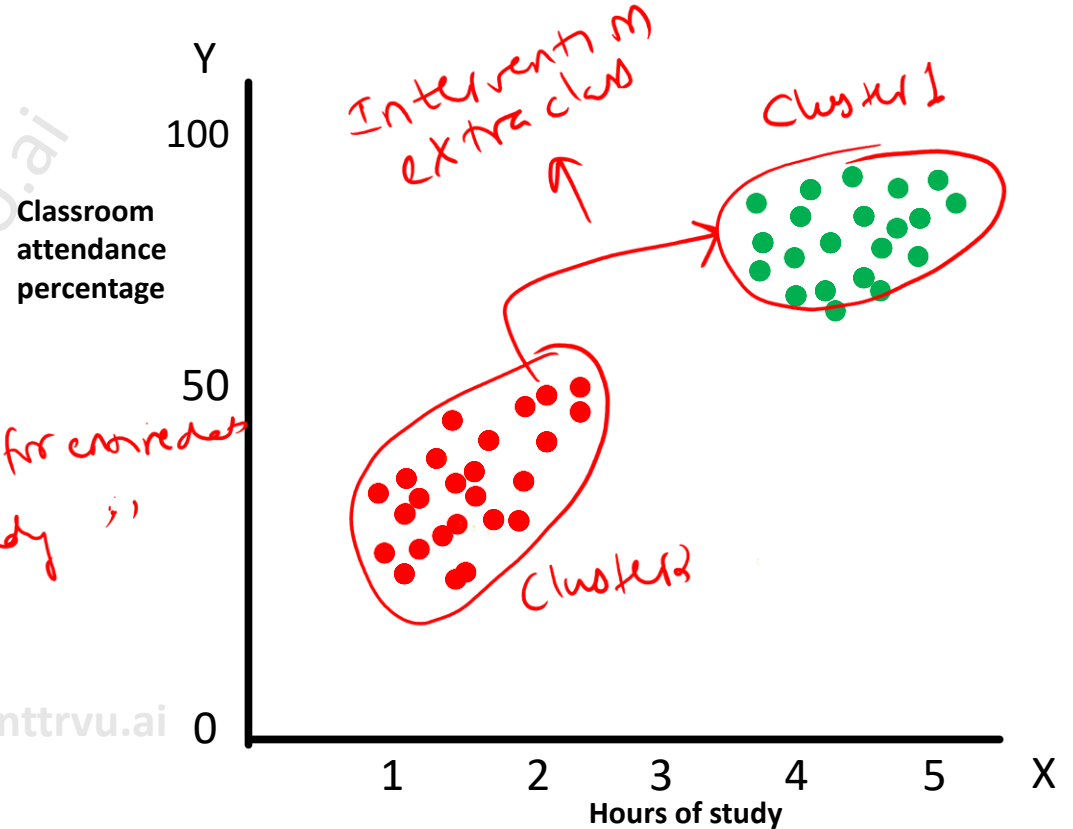
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What is clustering?

Cluster Profiling



Before Clustering



After Clustering

K-means clustering

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Cluster: is a group of similar

K-means clustering

↑
K number of clusters

data points

How to define this similarity b/w 2 data points?

- K-means clustering is method of clustering data points into k clusters (k is input to the algorithm)

k is an input to the algo; k is hyper-parameter

- Distance metric: Euclidean distance

$$R_k \rightarrow (82, 5.5)$$

$$R_i \rightarrow (80, 5)$$

$$R_j \rightarrow (50, 3)$$

$$d(R_i, R_j) = \sqrt{(80-50)^2 + (5-3)^2}$$

$$d(R_i, R_k) = \sqrt{(80-82)^2 + (5-5.5)^2}$$

- Algorithm Steps:

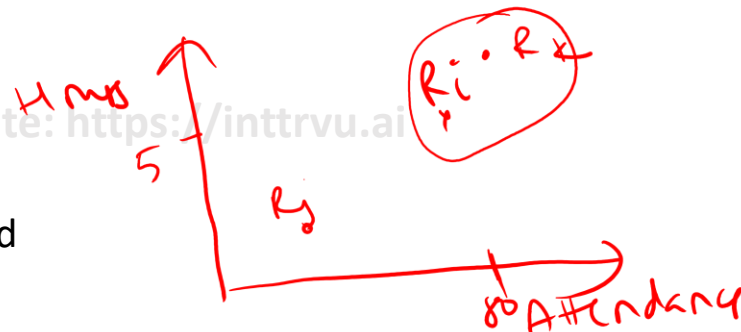
Initialization: Random initialization of k centroids

Assignment: Assign data points to nearest centroid based on Euclidean distance

Update: Update every cluster centroid by averaging all the data points in that cluster

Continue Assignment and Update until:

- Cluster centroids do not change further
- Specified number of iterations completed



$$d(R_i, R_k) < d(R_i, R_j)$$

R_i and R_k are similar to each other while R_j and R_i are dissimilar to each other

Steps of Algorithm

① Initialization (of what??)

We have to provide the number of clusters to make ; $k=3$

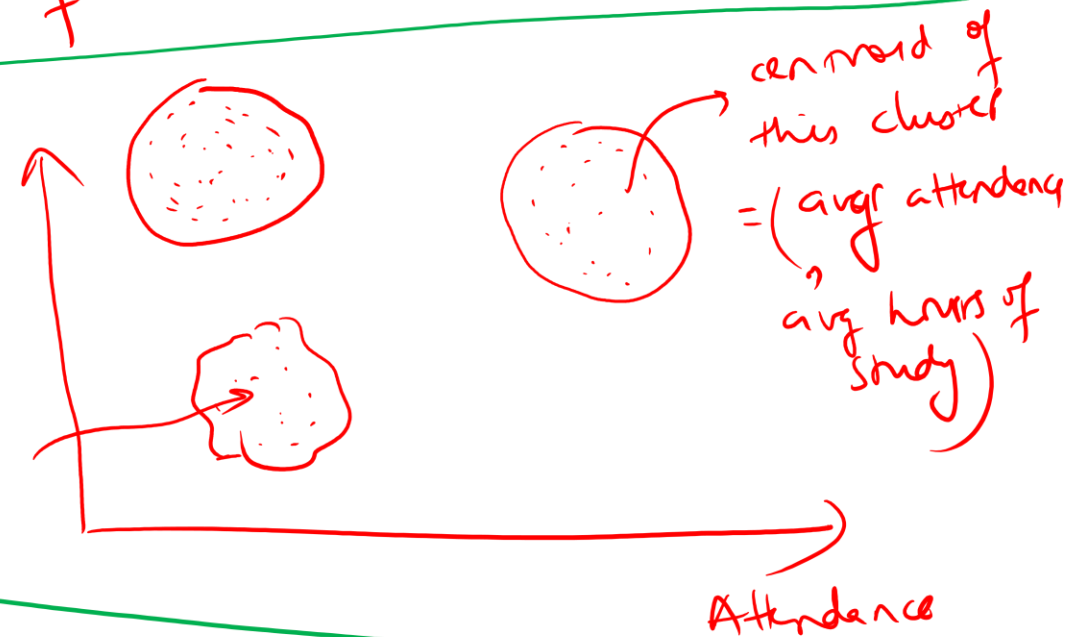
We have to initialize the centroids ; 3 centroids

.. .. k number of centroids

Centroid = (Avg. attendance, Avg. hours of study)

Definition/ calculation of centroid

Nothing to do with step 1



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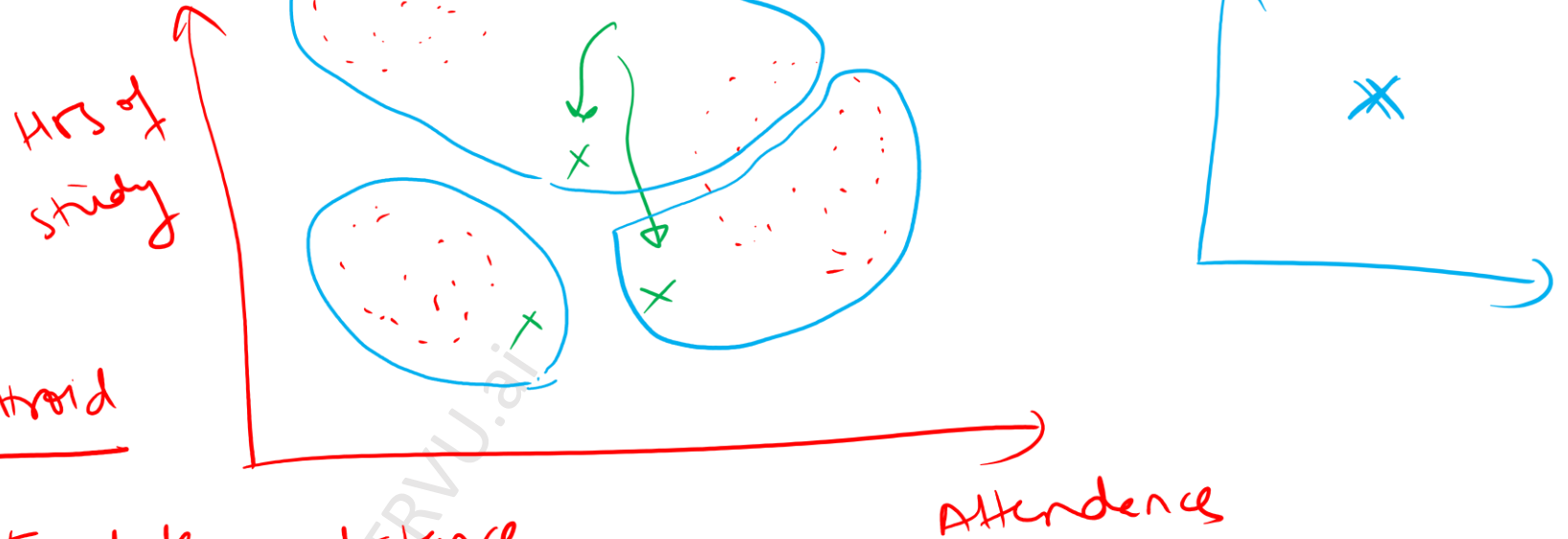
Step 2 Assignment

Assignment of
what?

Assignment of each data
point to the nearest centroid

I am going to use the Euclidean distance

Calculate distance of each point from every centroid . and assign data point
to that centroid which has lowest distance



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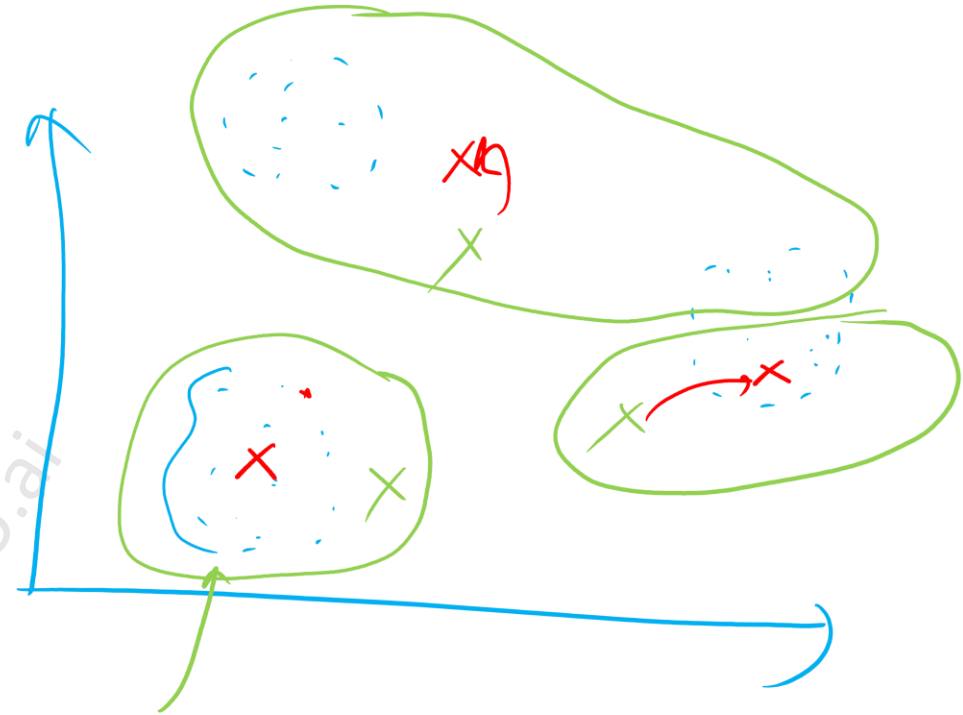
Step 3 Updation

Updation of centroids

to calculate the new centroid use
the averages

$(a_1, h_1), (a_2, h_2) \dots, (a_{10}, h_{10})$

$$\text{New centroid} = \left(\frac{1}{10} \sum a_i, \frac{1}{10} \sum h_i \right)$$



X→X

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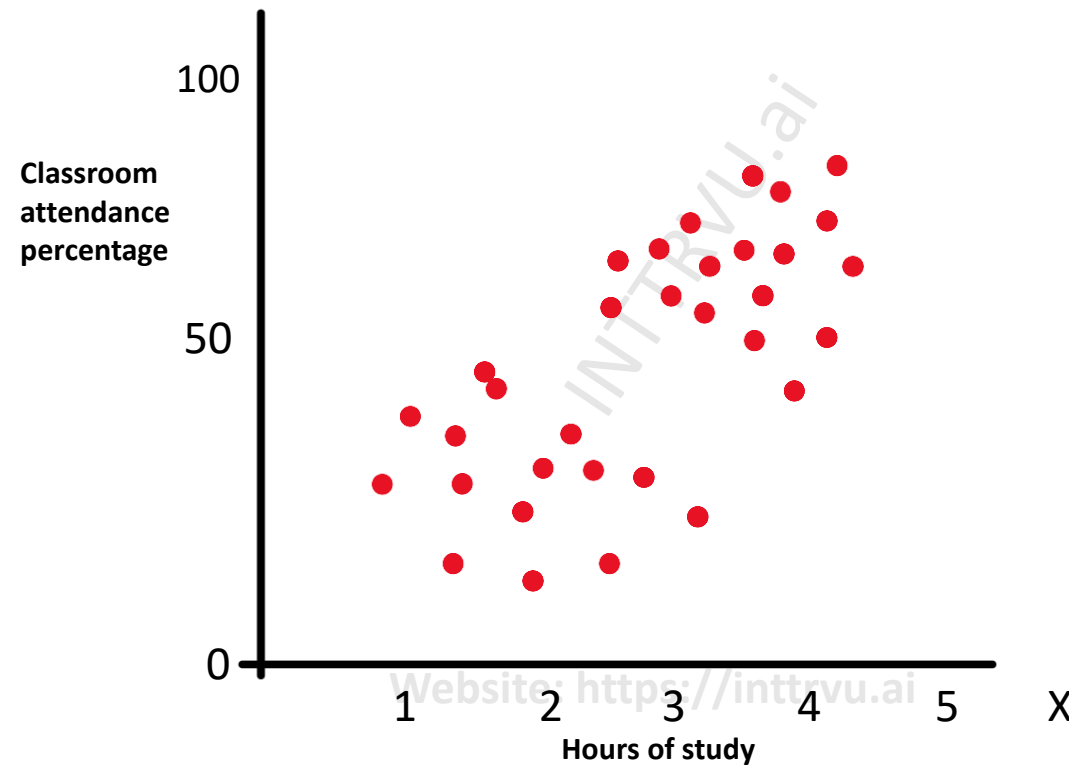
Now iterate for N number of times

Keep on repeating step 2 and step 3 till:

- ① Number of pre-decided iteration count has reached OR
- ② there is no change in assignment of the data points OR
- ③ there is no / very little change in centroid

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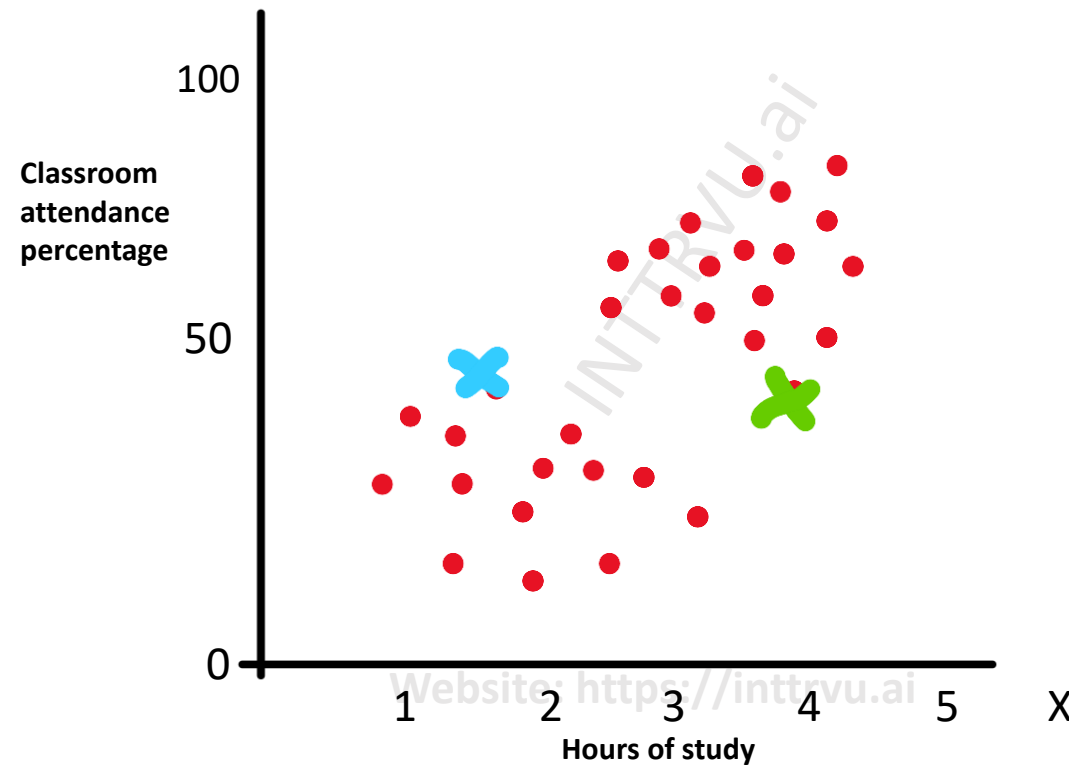
K-means clustering



$k = 2$

Dataset

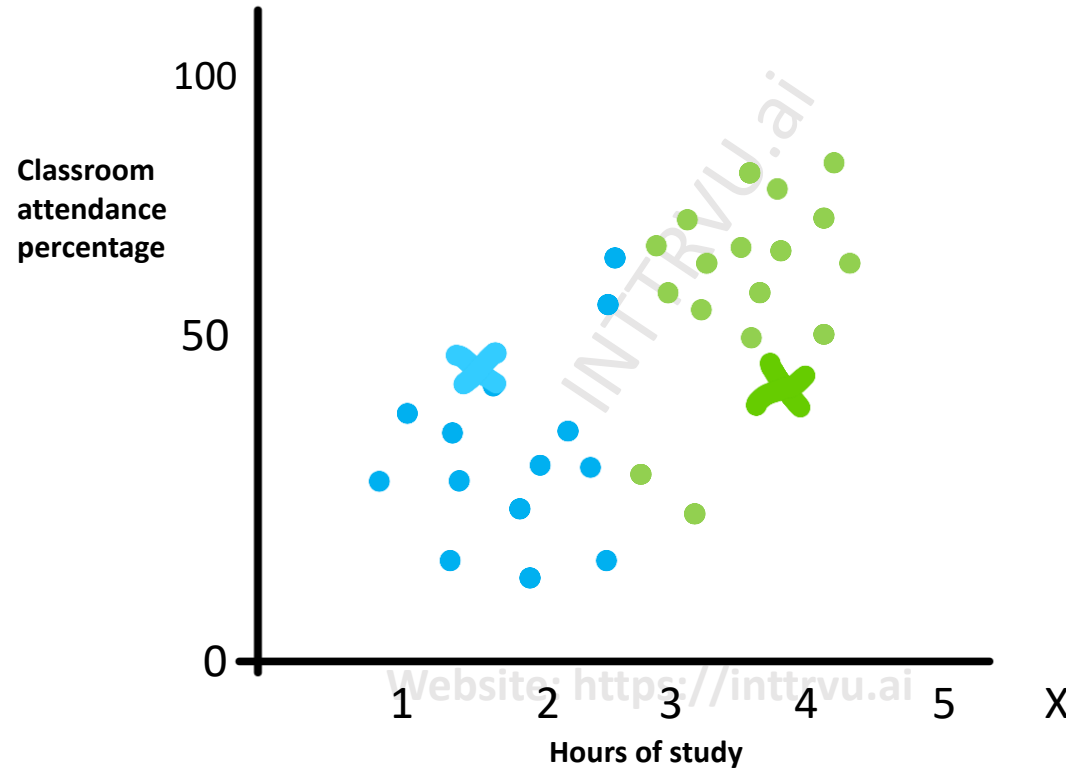
K-means clustering



Step 1

Initialization

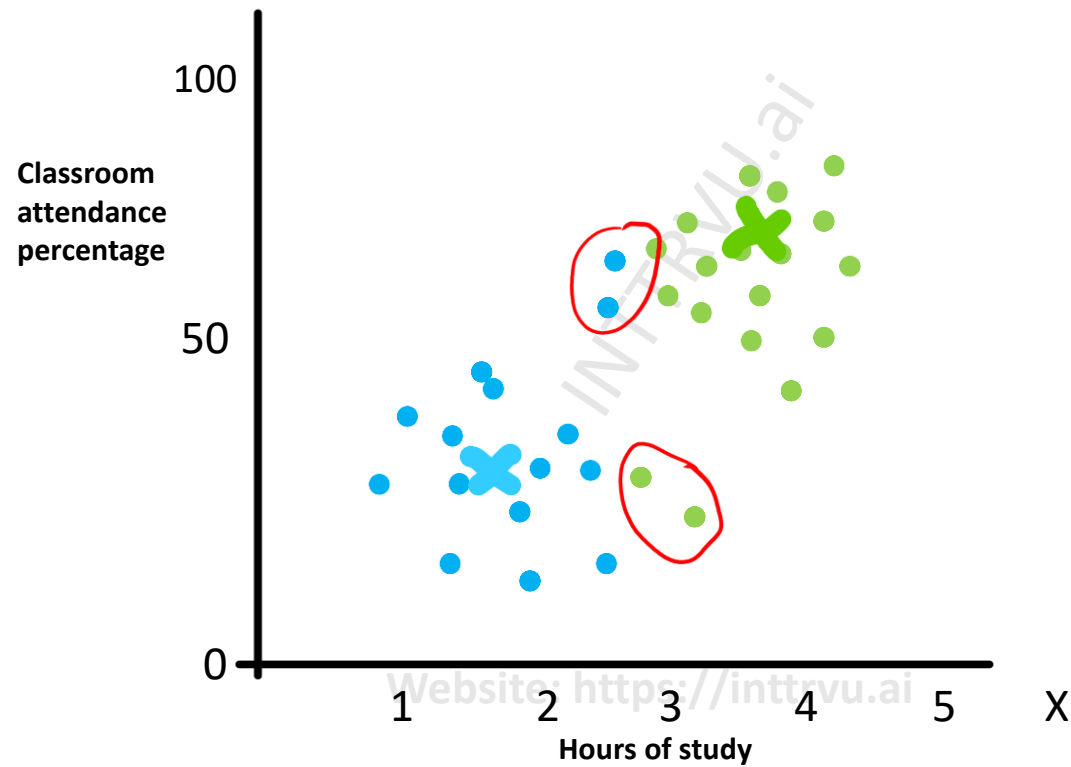
K-means clustering



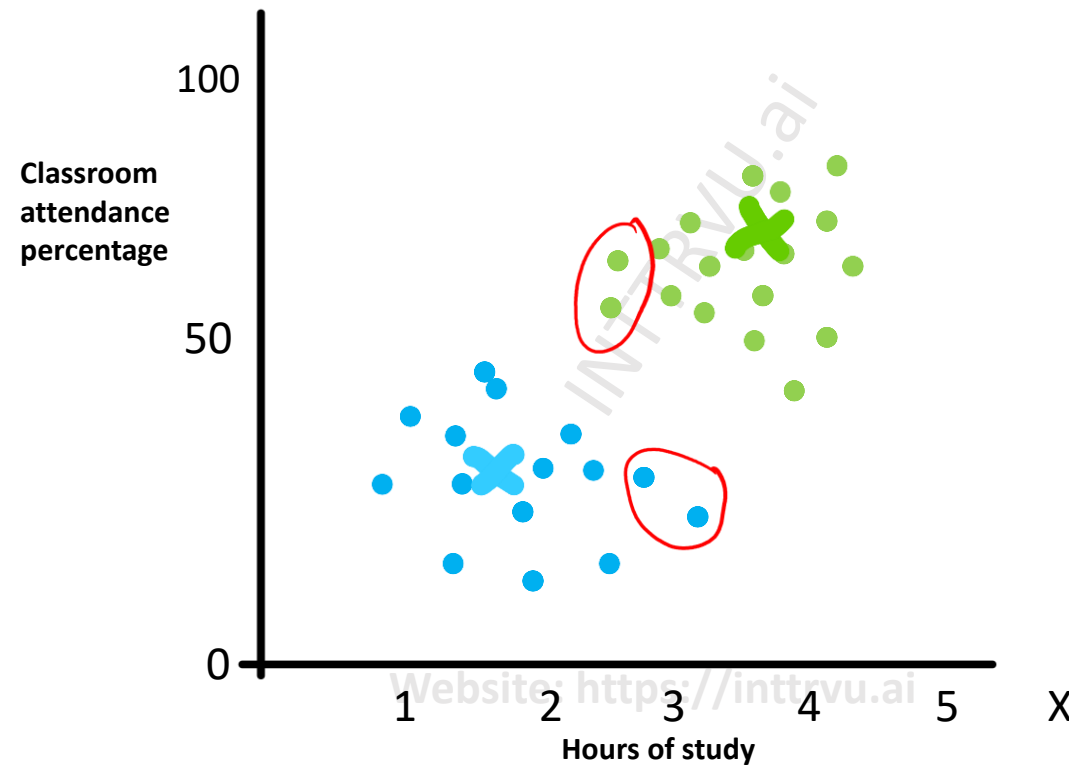
Step 2

Assignment

K-means clustering

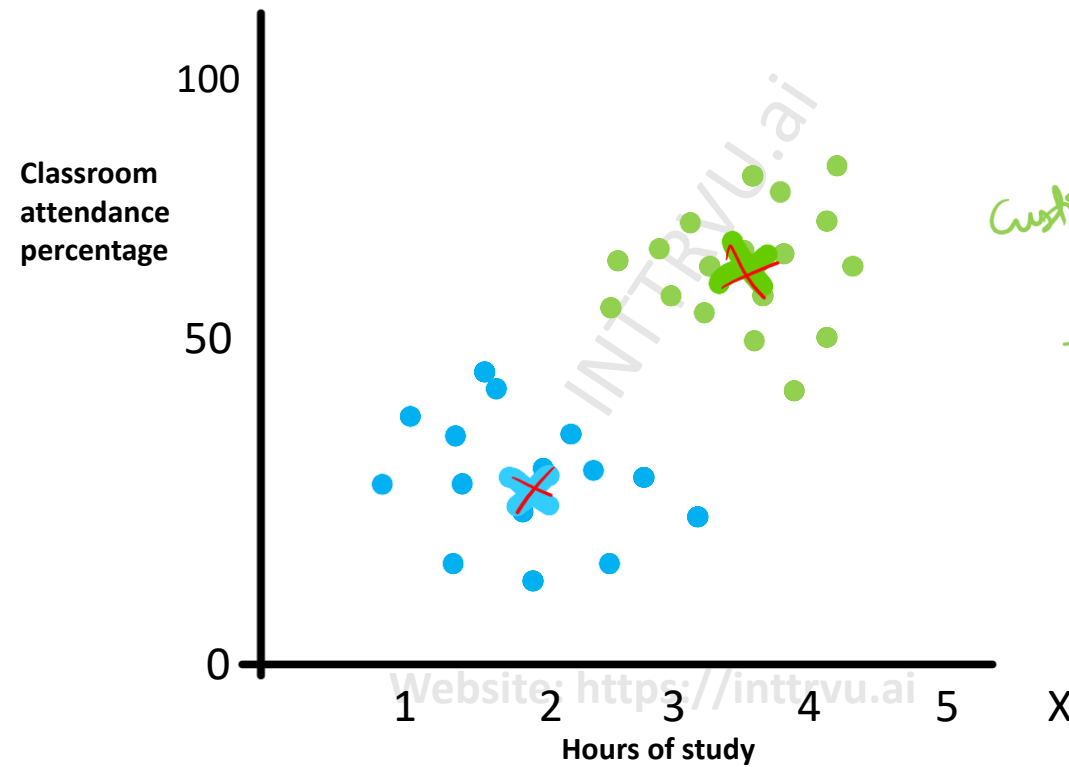


K-means clustering



Assignment

K-means clustering



Handwritten table structure for K-means clustering:

	F1	F2	F3	F4
Cluster 1				
C2				
C3				
C4				

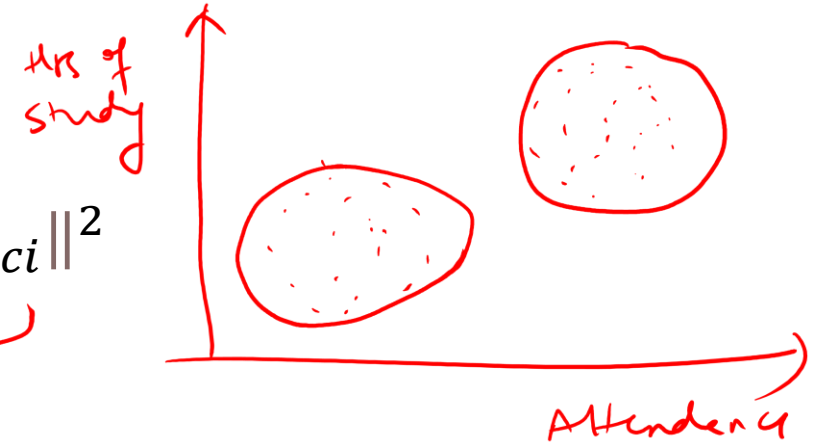
Update

Optimization Objective

- K-means optimization objective minimizes within-cluster variances (squared Euclidean distances)

- Optimization Objective: (cost function)

Minimize $\frac{1}{M} \sum_{i=1}^M \|x_i - \mu_{ci}\|^2$



M: Number of data points to be clustered

x_i : d dimensional data point

μ_{ci} : Centroid of cluster to which data point x_i is assigned (c is in range of 1 to k, where k is number of clusters)

Note: Optimization might converge at local minima

Cost Function

Residual Sum of Squares

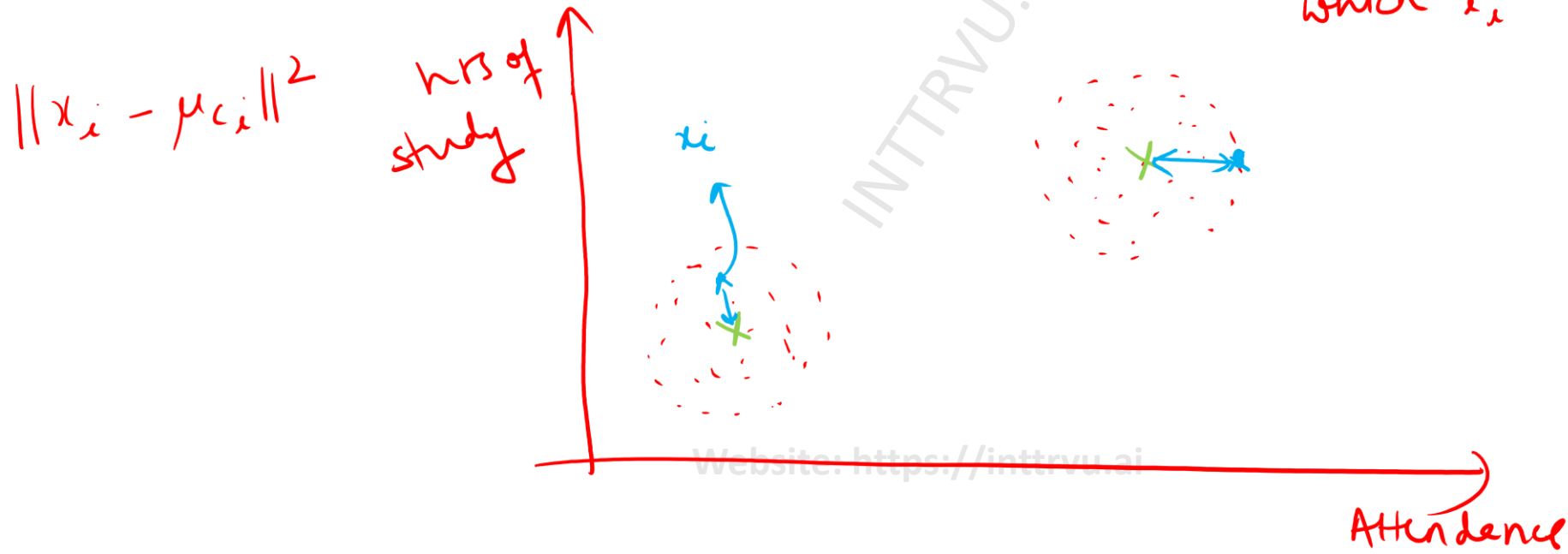
Minimize $\frac{1}{M} \sum_{i=1}^M \|x_i - \mu_{c_i}\|^2 = \text{RSS}$

$\|\cdot\|^2 \rightarrow$ Euclidean distance squared

$x_i \Rightarrow$ i^{th} data point / row

$M \Leftarrow$ total number of data points

$\mu_{c_i} \Rightarrow$ centroid of the cluster to which x_i belongs

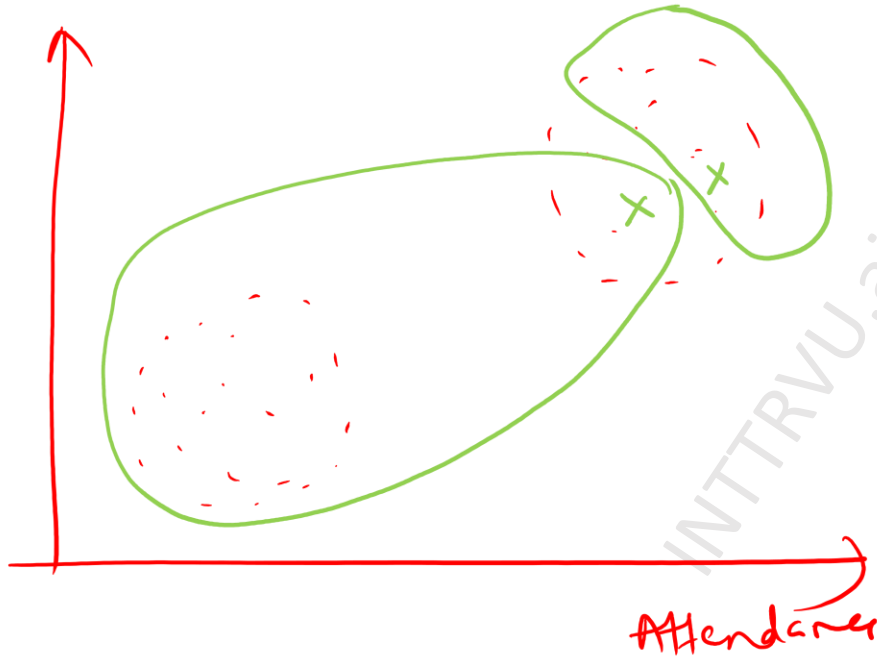


We have to minimize this cost function



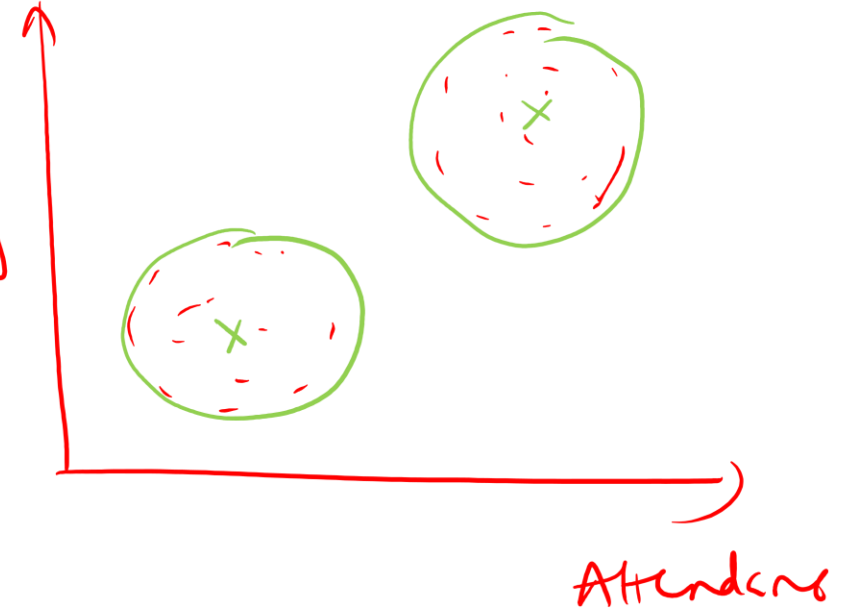
given the value of k
or given the number of clusters

hrs of
study



Scenario 1

hrs
of
study



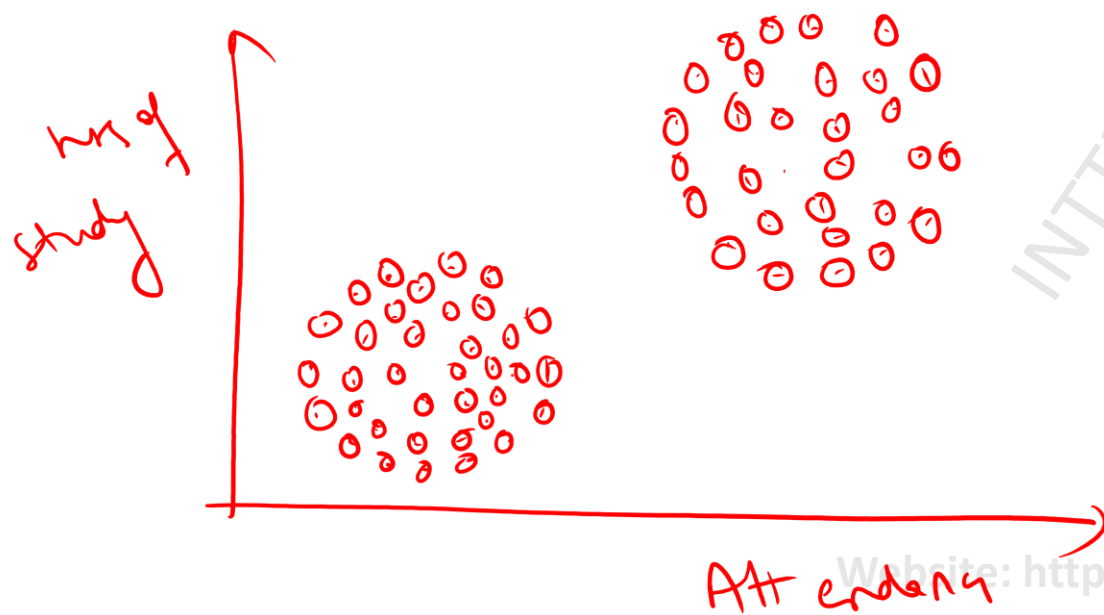
Scenario 2

Website: <https://inttrvu.ai> $RSS_2 < RSS_1$

What is going to happen if we don't specify value of k?

$$RSS = \frac{1}{n} \sum \|x_i - \mu_c\|^2 \quad \checkmark$$

Objective is to minimize RSS



$$K = 100$$

100 clusters

In every cluster, there will be only 1 point

⇒ Every point will be its own centroid

$$\Rightarrow RSS = 0$$

Need of Feature Scaling

- In our example
 - 'Hours of study' feature is in the range of 0 to 5
 - 'Classroom attendance' feature is in the range of 0 to 100
- As k-means clustering uses Euclidian distance metric it will suffer if we have features with different scales
- In this example if we apply clustering without scaling then Classroom attendance feature will drive all the clustering with very low importance to hours of study

Solution:

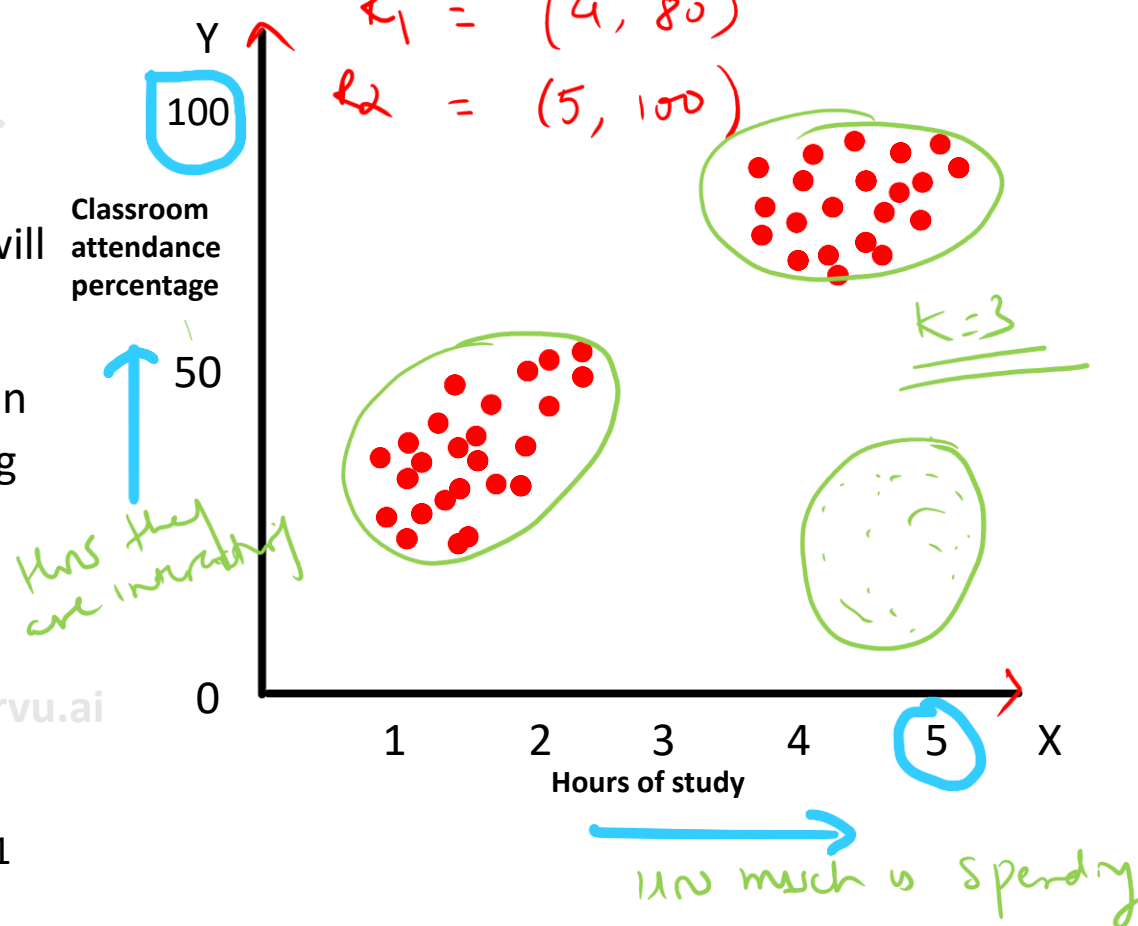
Scale features to equal scale wherever possible

In this example both the features can be scaled to 0 to 1

$$d(R_1, R_2) = \sqrt{(5-4)^2 + (100-80)^2}$$
$$= \sqrt{1^2 + 20^2} = \sqrt{401} \approx (20)$$

$$R_1 = (4, 80)$$

$$R_2 = (5, 100)$$



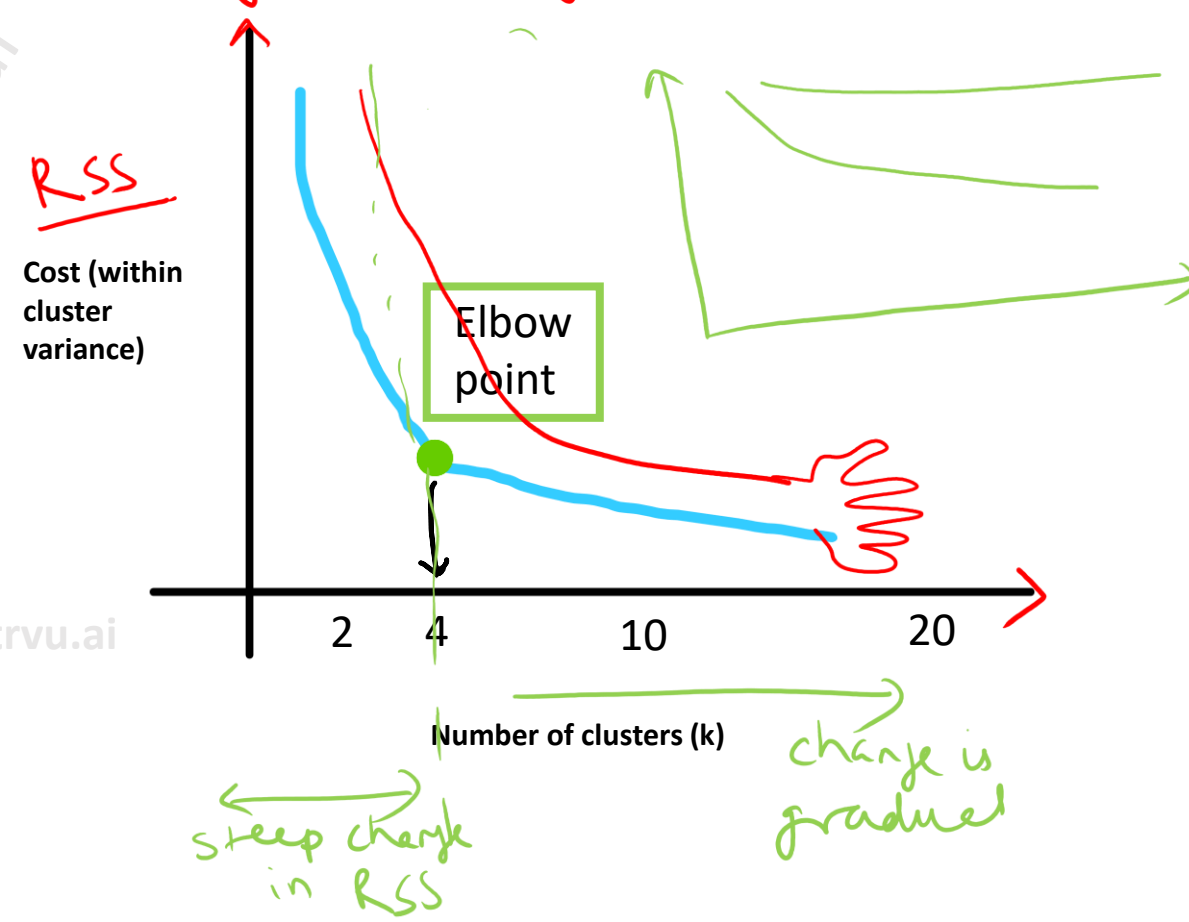
How to decide value of k?

the value of k is decided based on some experiments.

$k = [2, 3, 4, 5, \dots, 19, 20]$

Run k means clustering for each and every value of k in this range

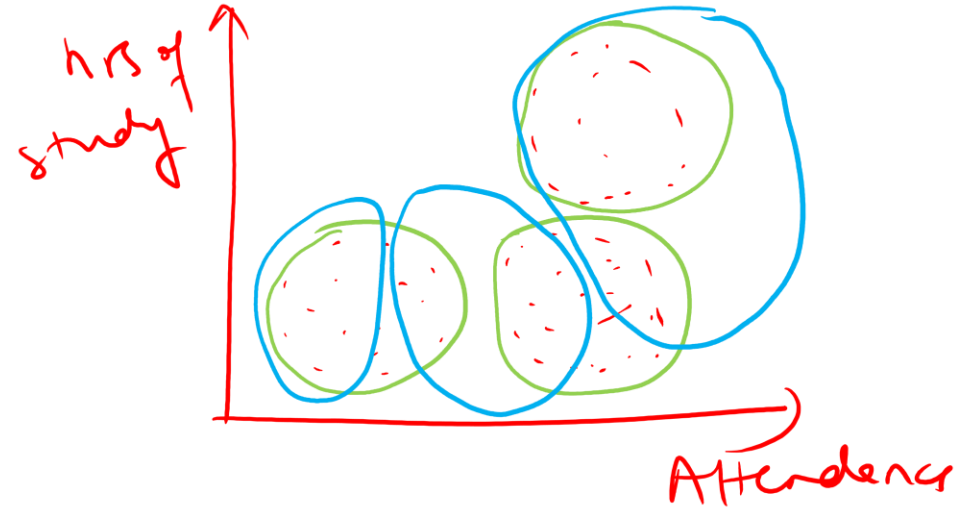
- Run k-means for various number of clusters
- Choose number of clusters which represent elbow point on cost curve
- **Key points to remember:**
 - Elbow point doesn't guarantee best results
 - In many cases a clear elbow point won't be found
 - Choose number of clusters based on business objective and evaluate clustering results based on the same
 - Understand how clustering results would be used
 - Check if stakeholders have any idea about number of groups/clusters they are expecting



Effect of Initialization of centroids on k-means

Method 1 Run k-means clustering multiple times say 10 times

1	$k=3$	RSS_1 ✓
2	$k=3$	RSS_2
⋮		
10	$k=3$	RSS_{10}



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Method 2 K-Means ++ [enhanced version of k-Means]

only step 1 is different ; step 2 and 3 are exactly same

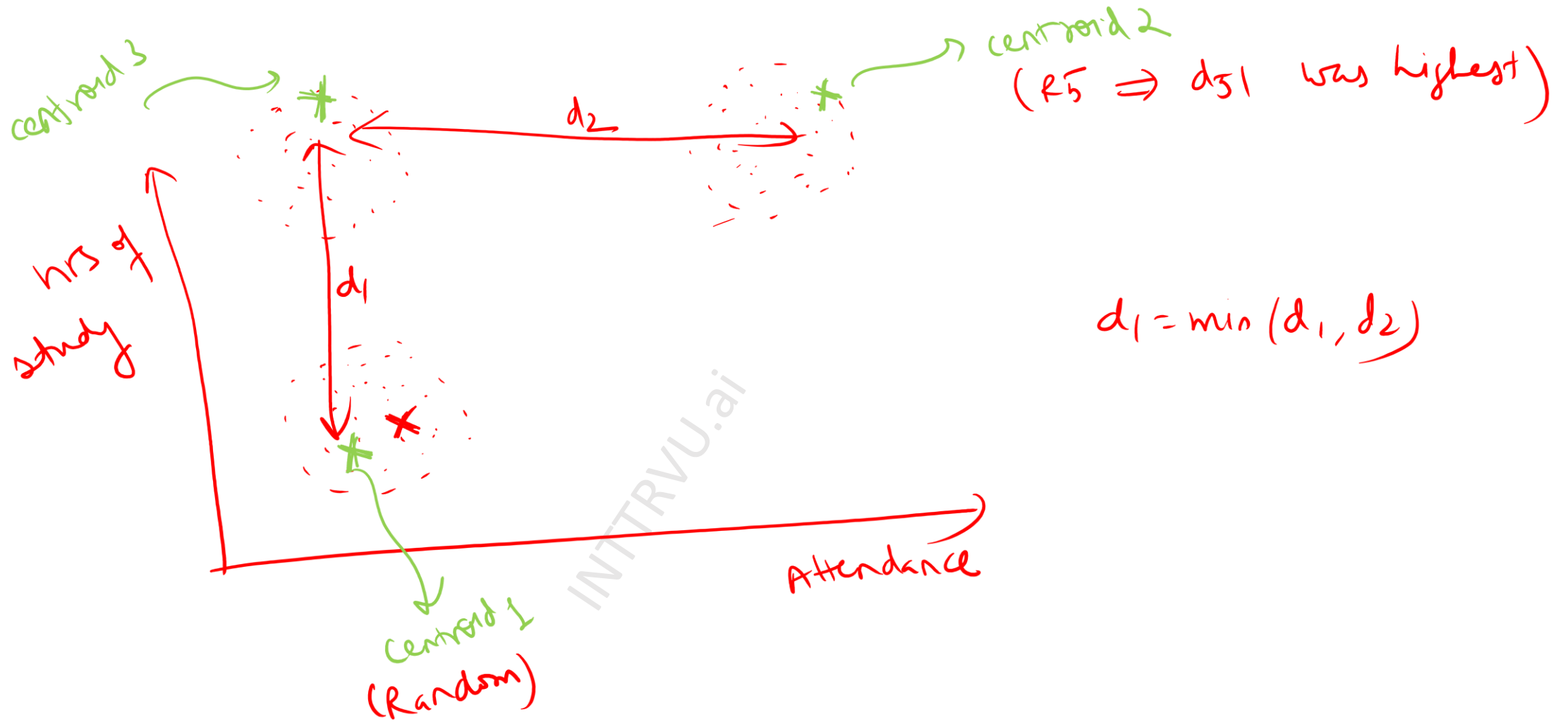
let say, we have to initialize 3 centroids $k=3$

Step 1 Choose 1st centroid randomly

Step 2 Now choose 2nd centroid which is farthest away from centroid 1

Step 3 choose 3rd centroid which is farthest away from centroid 1 and 2.

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① First centroid will be chosen randomly

② for second centroid

	F_1	F_2	Distance from 1 st centroid
R1			d_{11}
R2			d_{21}
R3			d_{31}
R4			d_{41}
R5			d_{51}
R6			d_{61}

second
centroid

← ✓ R5

choose that row as second centroid which has highest distance value
 $\Rightarrow d_{51}$ is the highest $\Rightarrow R5$ is the second centroid

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③ For third centroid

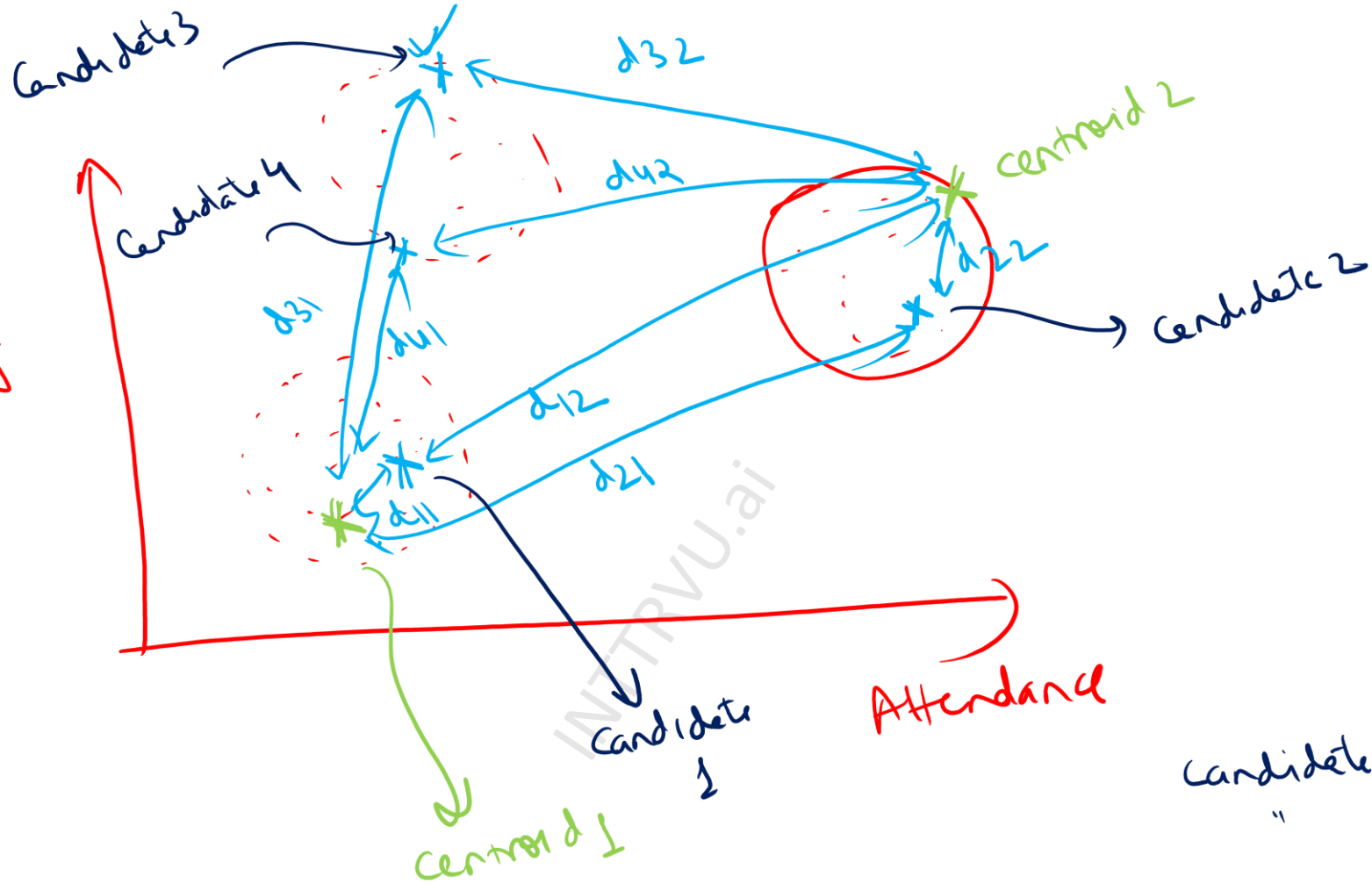
	F1	F2	Distance from centroid 1	Distance from centroid 2	d _{min}
R1			d ₁₁	d ₁₂	min(d ₁₁ , d ₁₂)
R2			d ₂₁	d ₂₂	min(d ₂₁ , d ₂₂)
R3			d ₃₁	d ₃₂	min(d ₃₁ , d ₃₂)
R4			d ₄₁	d ₄₂	min(d ₄₁ , d ₄₂)
R5			d ₅₁	d ₅₂ = 0	min(d ₅₁ , d ₅₂)
R6			d ₆₁	d ₆₂	min(d ₆₁ , d ₆₂)

highest value amongst the ~~columns~~ rows in this column

choose that row as third centroid for which d_{min} is highest

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hrs of study

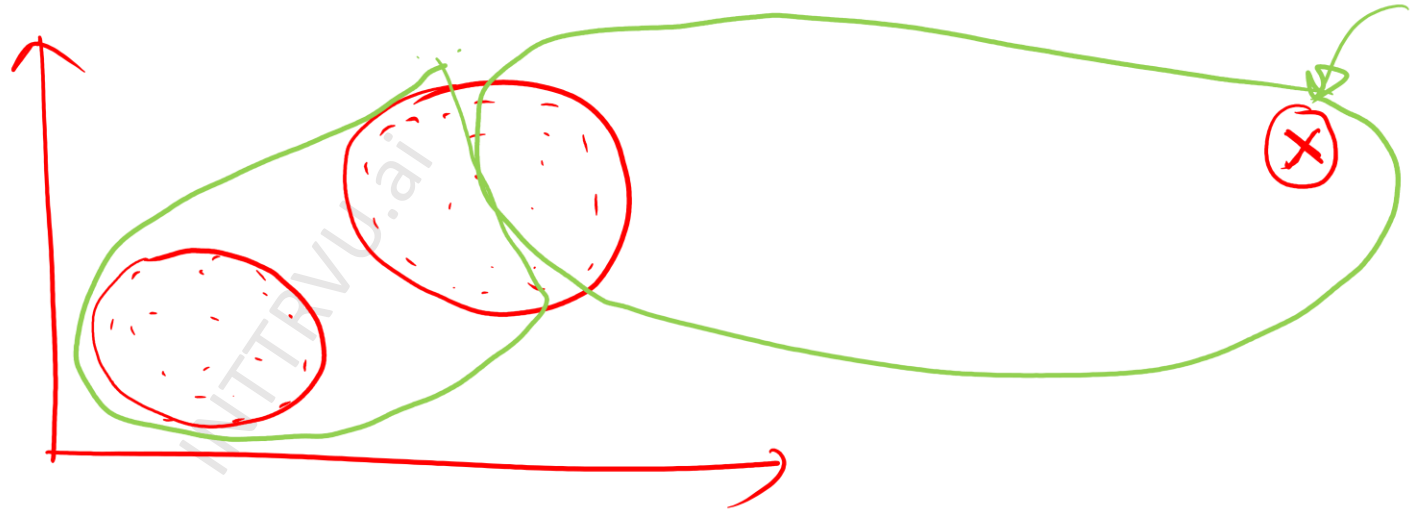


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Candidate 1	d_{11}	
" 2	d_{22}	
" 3	d_{31}	✓
" 4	d_{41}	

Problems with K-means

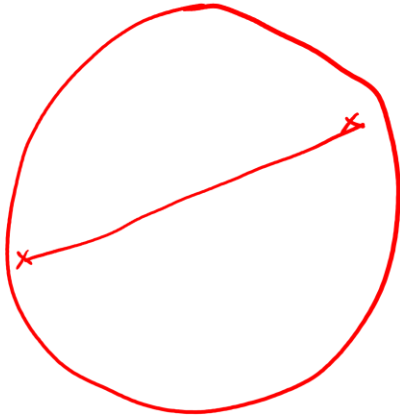
- ① You have to provide the value of K
- ② K-means Algo doesnot deal with outliers automatically



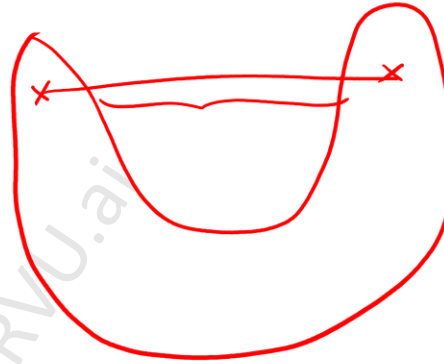
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③ It can only make convex shapes | convex shaped clusters.

What is a convex shape??



Convex shape



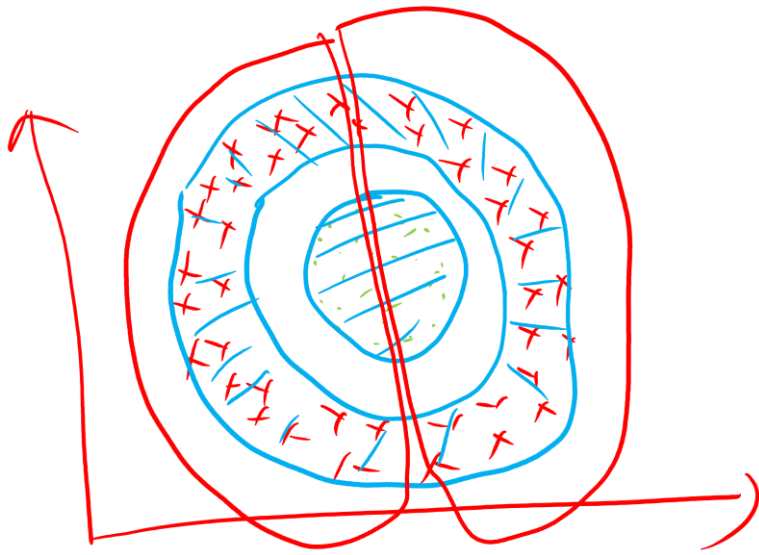
Concave Shape



Concave shape



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④ We can only use Euclidean distance

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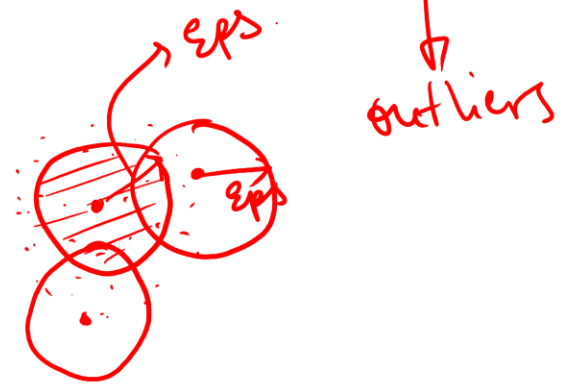
It is a clustering algorithm

Density based

spatial clustering for applications with noise

Algorithm works on the concept density.

Density in the feature space



DBSCAN Clustering

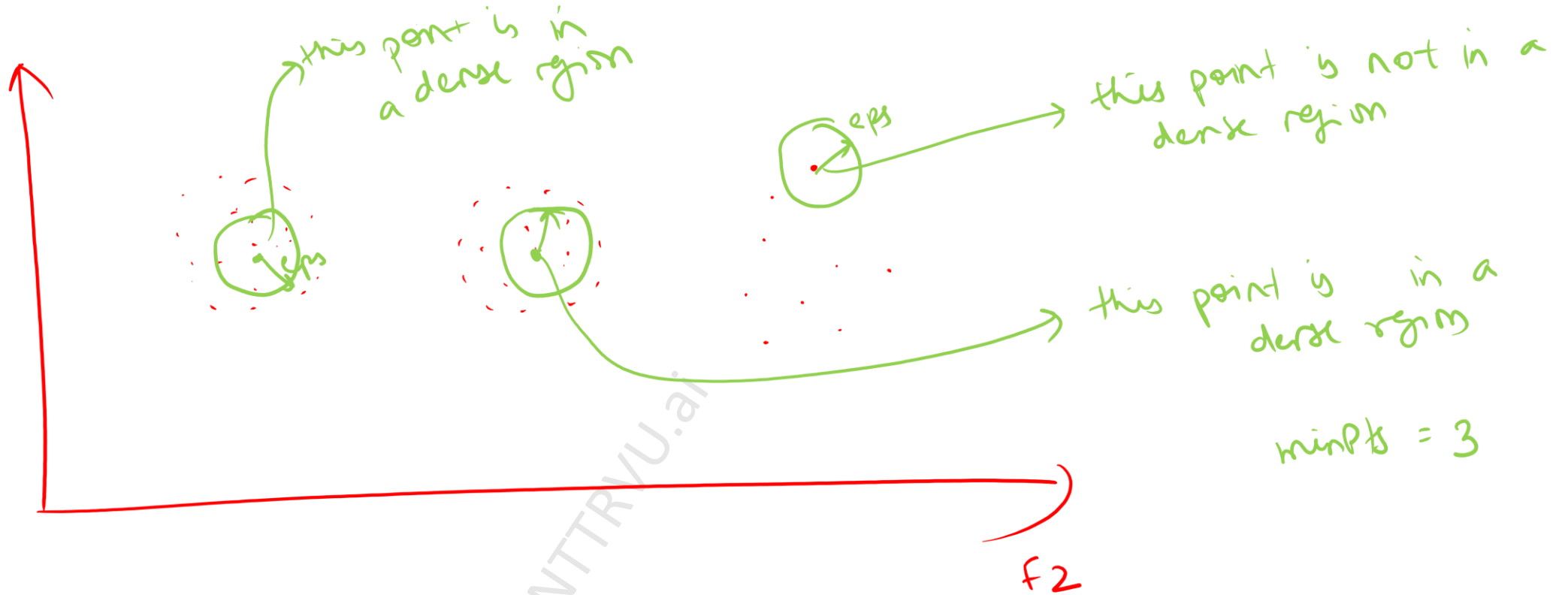
How is density? ✓
How the outliers are handled? ✓
How the outliers are defined? ✓

In terms of number of neighbors.

✓ ϵ (eps)
✓ min pts ()

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F1



eps $\rightarrow$ to define the region
minPts $\rightarrow$ to define the threshold for number of neighbors.
If number of neighbors $>$ minPts $\Rightarrow$ dense region
else $\Rightarrow$ not dense region.

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DBSCAN Clustering

Higher the density $\Rightarrow$ Algo will group the points together.

- DBSCAN stands for Density-based spatial clustering of applications with noise (DBSCAN)

- It groups together points that are closely packed together (points with many nearby neighbors), and points that lie alone in low-density regions are marked as outliers

A dense region is a region with a lot of points

- Distance metric: Euclidean, Cosine, Hamming, Mahalanobis etc.

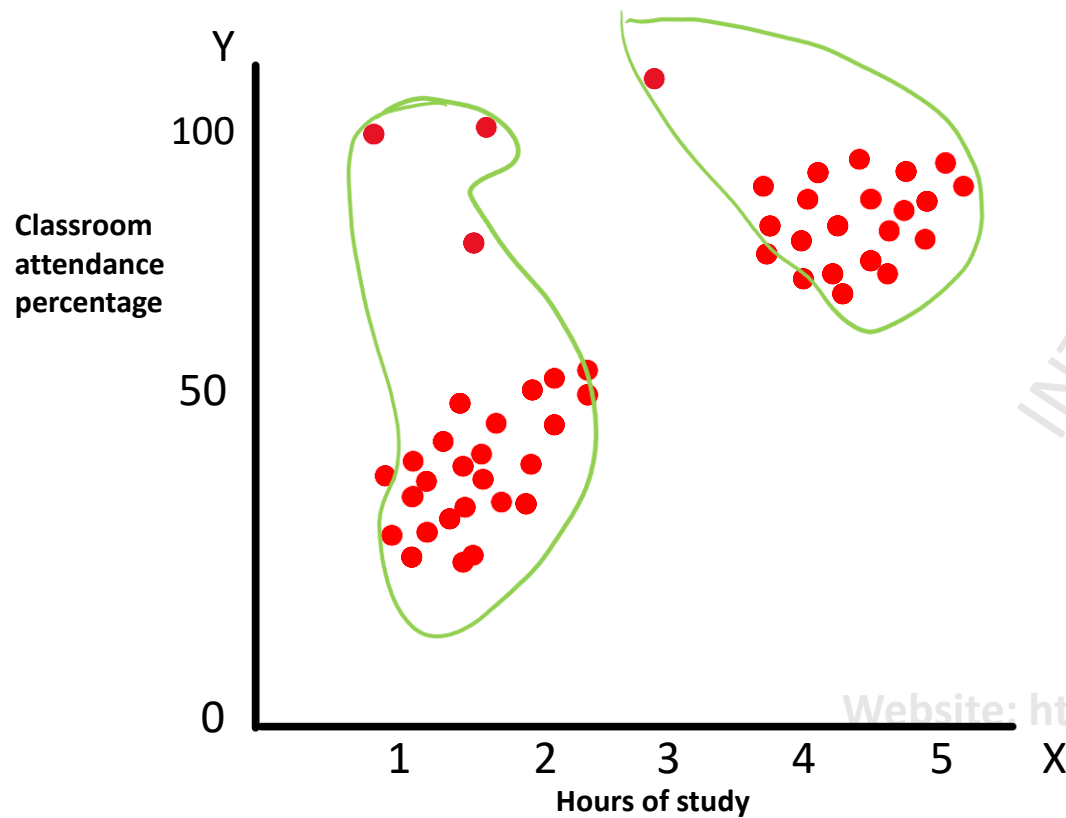


- **Parameters:**

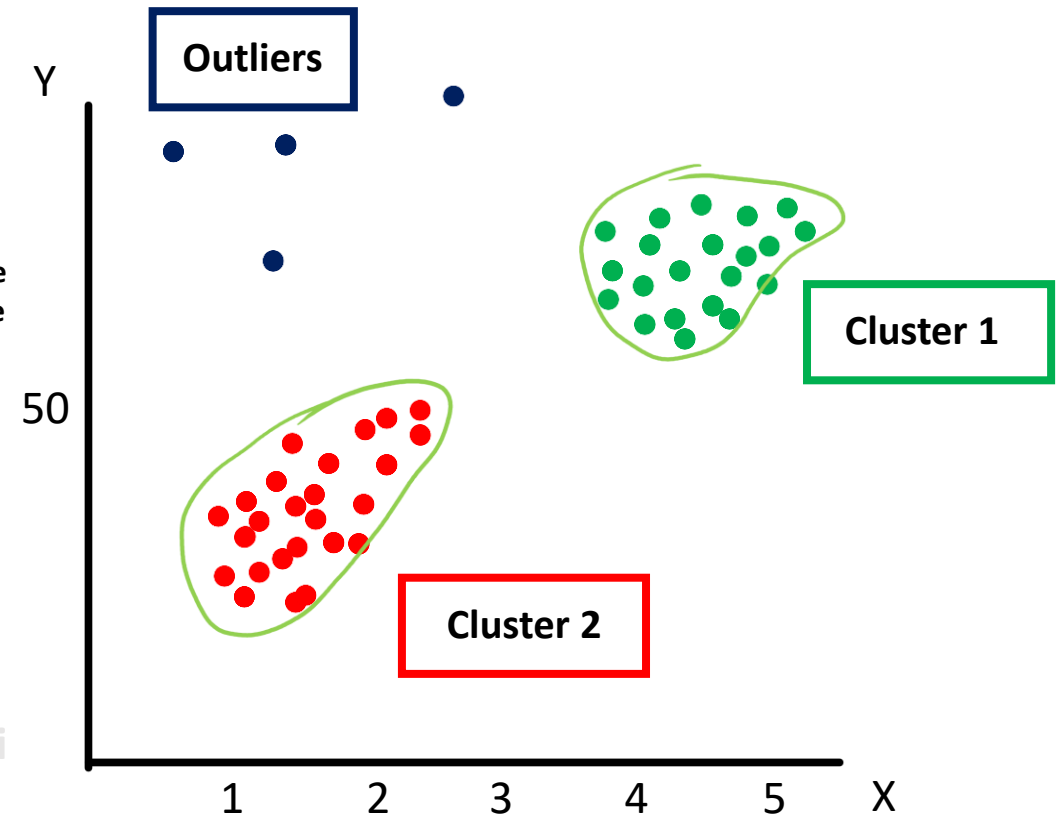
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- ✓ ϵ (eps): distance threshold to consider a point as neighbor
- ✓ minPts: Minimum number of points needed in neighborhood to form a cluster

DBSCAN Clustering



Before Clustering



After Clustering

we give labels to each points : core point, non-core point and outlier

DBSCAN Clustering Algorithm

A core point is a point having $\geq \text{minPts}$ as neighbor

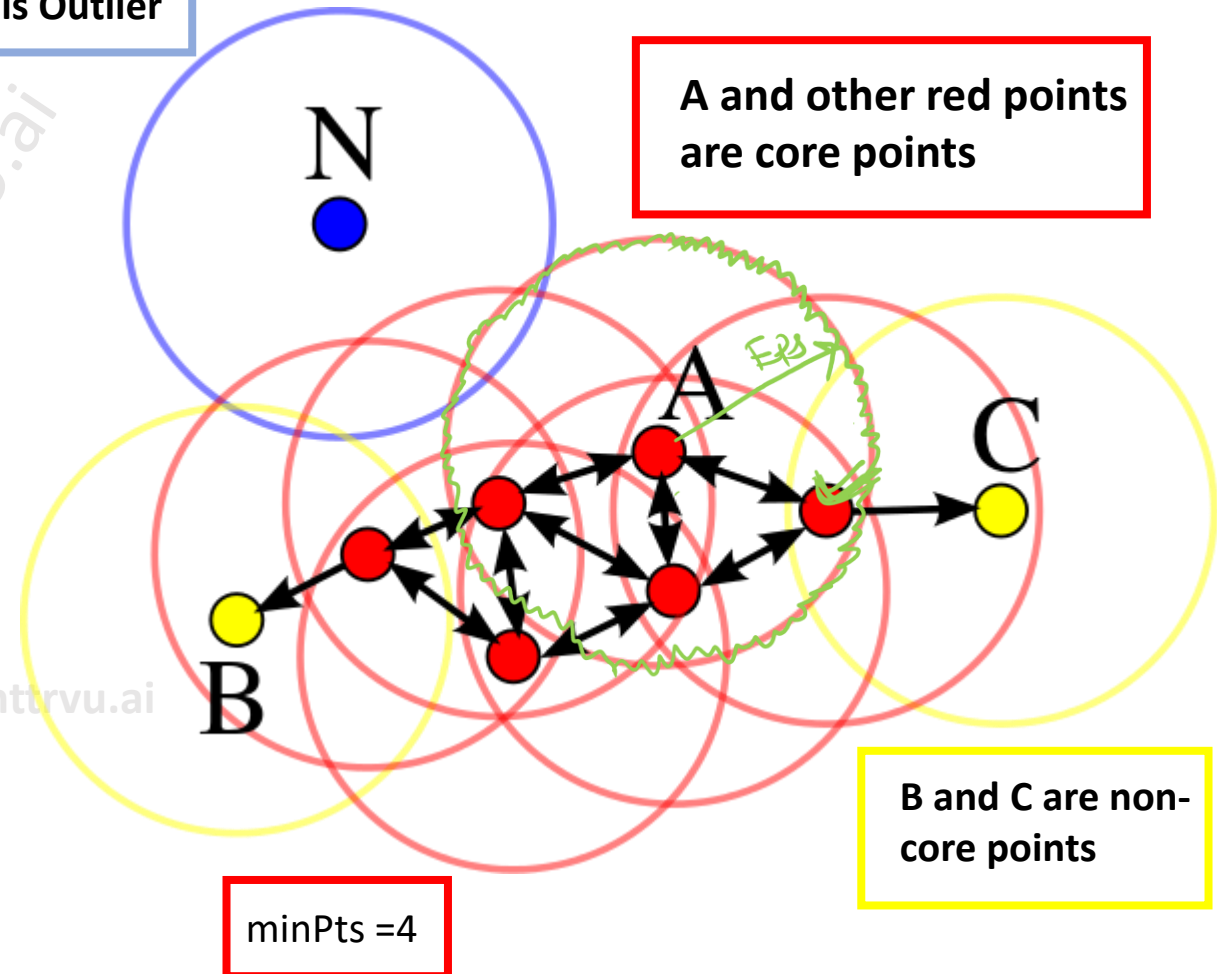
A non core point is a point having $< \text{minPts}$ neighbor but it has a core point as a neighbor

• Steps:

- Find the points around (neighbors) every point which are within distance ϵ (eps)
- Label the points as core points if they have more than minPts neighbors (including the point itself)
- Connect core points (consider them as belong to same cluster) if they are within distance ϵ (eps) from each other
- Assign each non-core point to a nearby cluster if any point in cluster is within distance ϵ (eps) neighbor, otherwise assign it to noise (outliers)

N is Outlier

A and other red points are core points



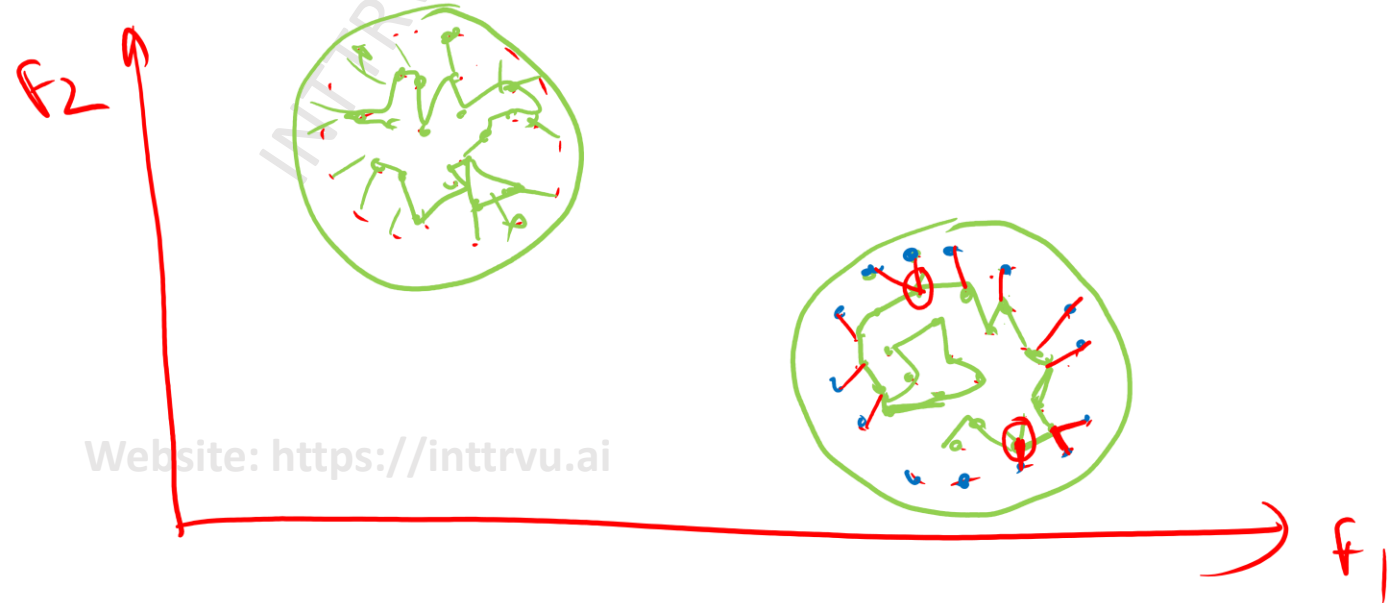
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Three Labels

- ① Core point : a point is a core point if it has ~~more than~~ at least minPts as neighbor
- ② Non core point : If it has less than minPts as neighbor and at least 1 core point as neighbor
- ③ Outlier : Less than minPts neighbor and no core point as neighbor

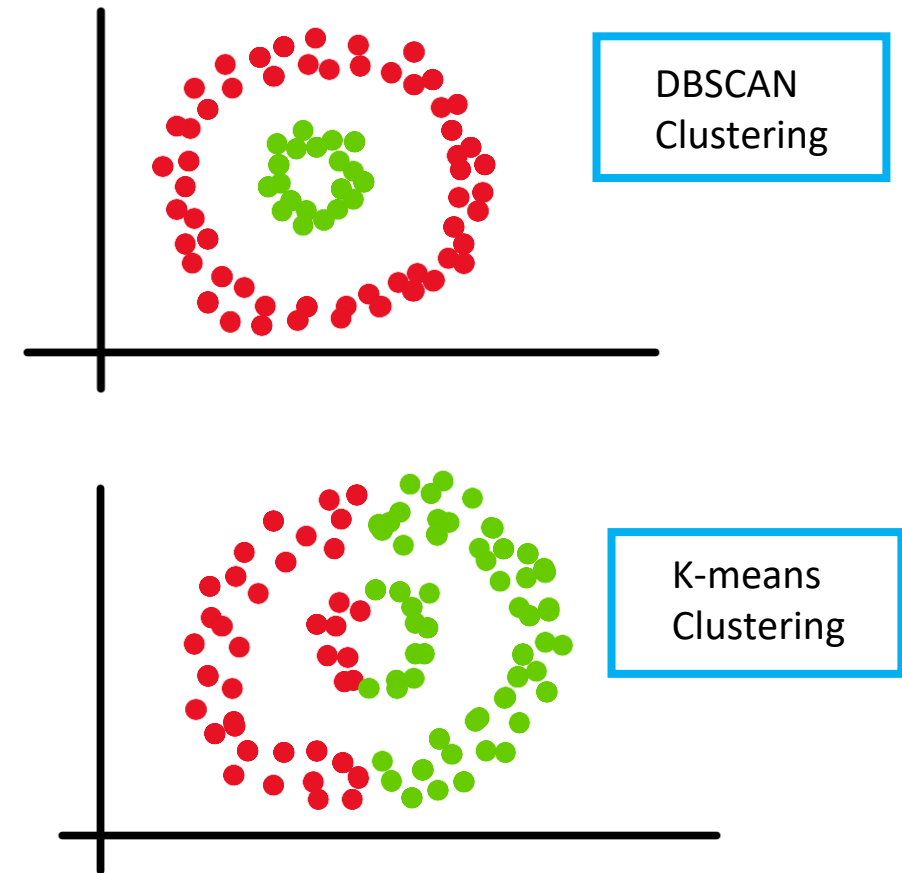
ϵ ps, minPts



Advantages of DBSCAN Clustering

- If the data domain is well understood then ϵ and minPts can be chosen appropriately for clustering, no need to specify number of clusters
- Works well even for clusters with arbitrary shape
- It can identify and separate outliers instead of just adding them to nearest cluster
- We can use distance metrics other than Euclidean distance

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DBSCAN

Eps, minPts

1
2

4

1 → 1.5

1 → 0.5

1 → 2

1 → 1.1

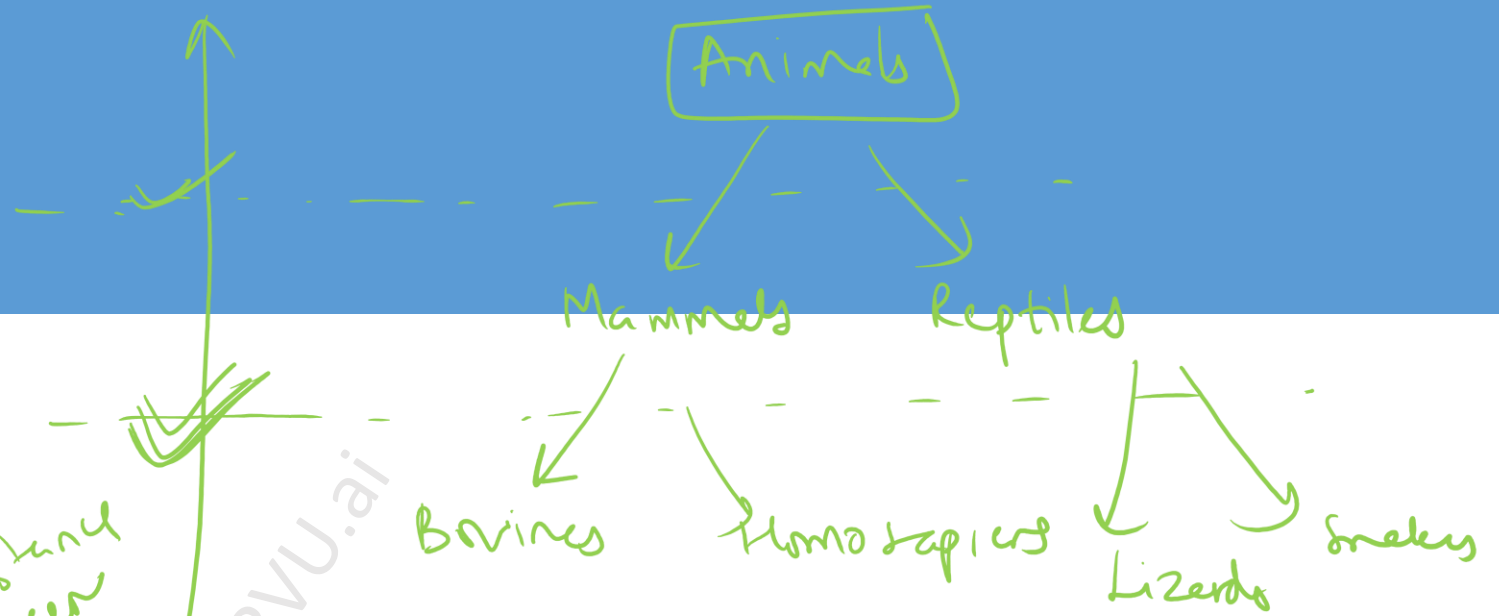


distance
between
genes

Hierarchical Clustering

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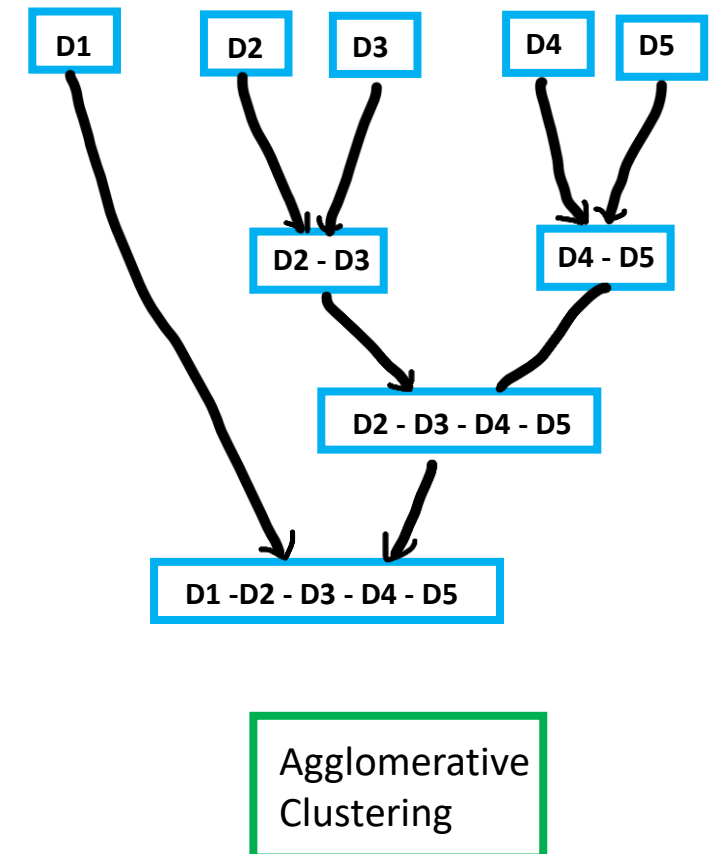
entire population
on earth



Hierarchical Clustering

- Hierarchical Clustering method builds a hierarchy of clusters
- Types of Hierarchical Clustering:
 - ✓ • **Agglomerative**: Each observation starts in its own cluster, and pairs of clusters are merged based on distance metric and 'linkage criteria'
 - ✓ • **Divisive**: All observations start in one cluster, and splits are performed recursively based on distance metric and 'linkage criteria'
- Distance metric: Euclidean, Cosine, Manhattan etc.

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Let's take an example of medical documents : 100 documents

distance metrics
↑

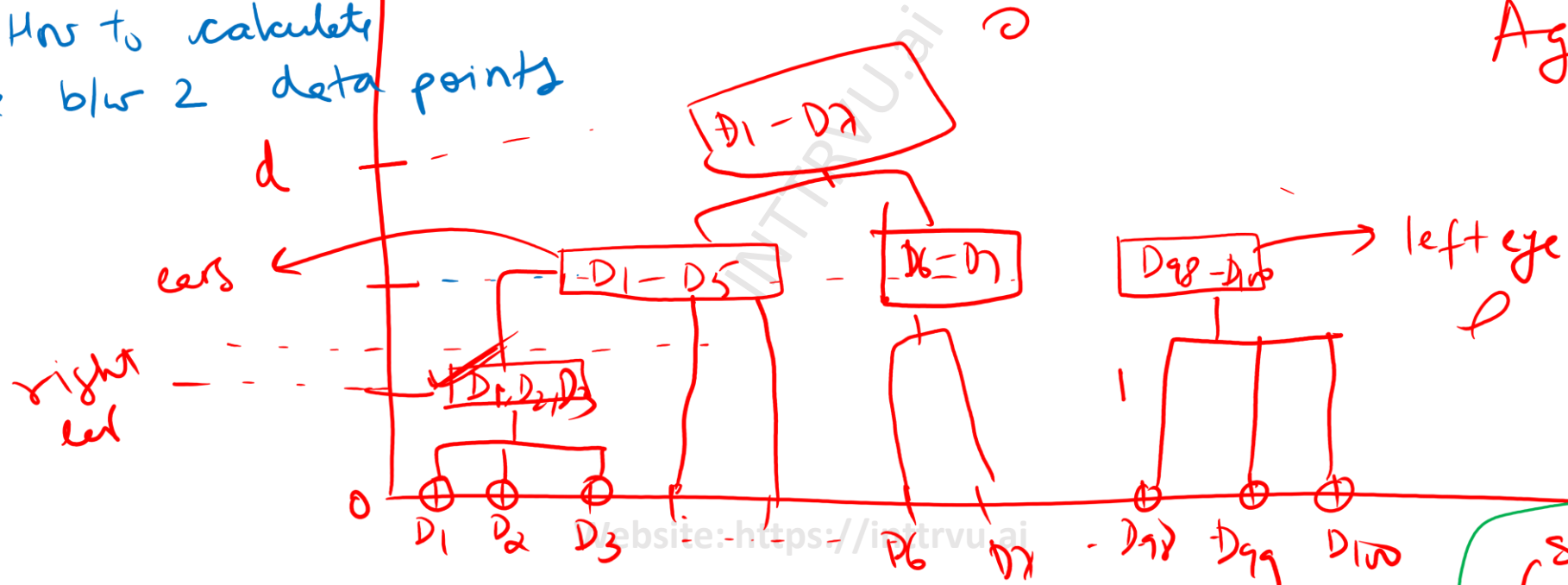
distance threshold.

only 1 cluster

starting point for divisive clustering

How many clusters here??

Issue 1 How to calculate distance b/w 2 data points



Agglomerative clustering

Issue 3 : How to calculate distance between a point and a cluster

Issue 3 How to calculate distance b/w 2 clusters

starting point for Agglomerative

Linkage criteria - Single Linkage

Q - What is the distance between C1 and C2 ??
= The distance between closest points of cluster

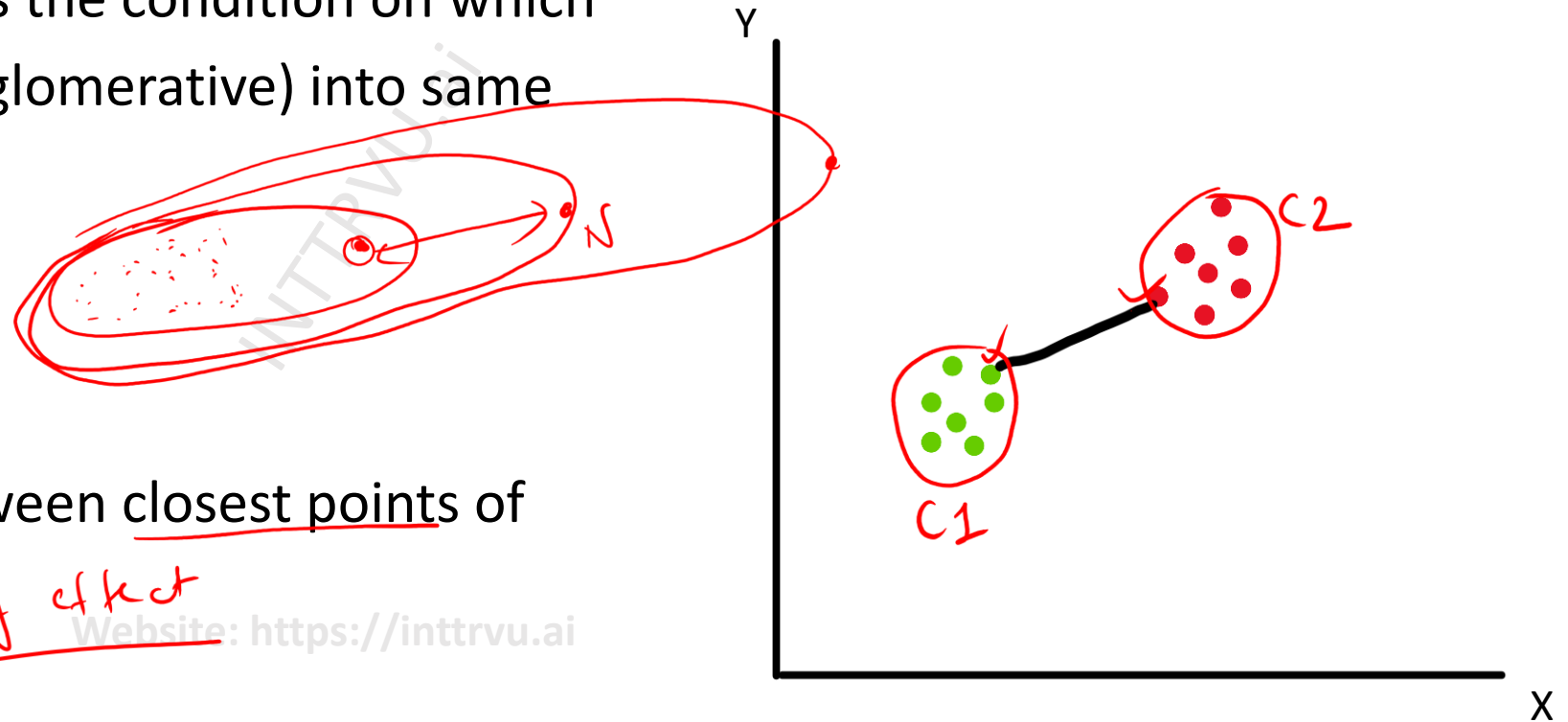
- Linkage Criteria decides the condition on which points are merged (agglomerative) into same cluster

- **Single Linkage:**

It is the distance between closest points of cluster

chaining effect

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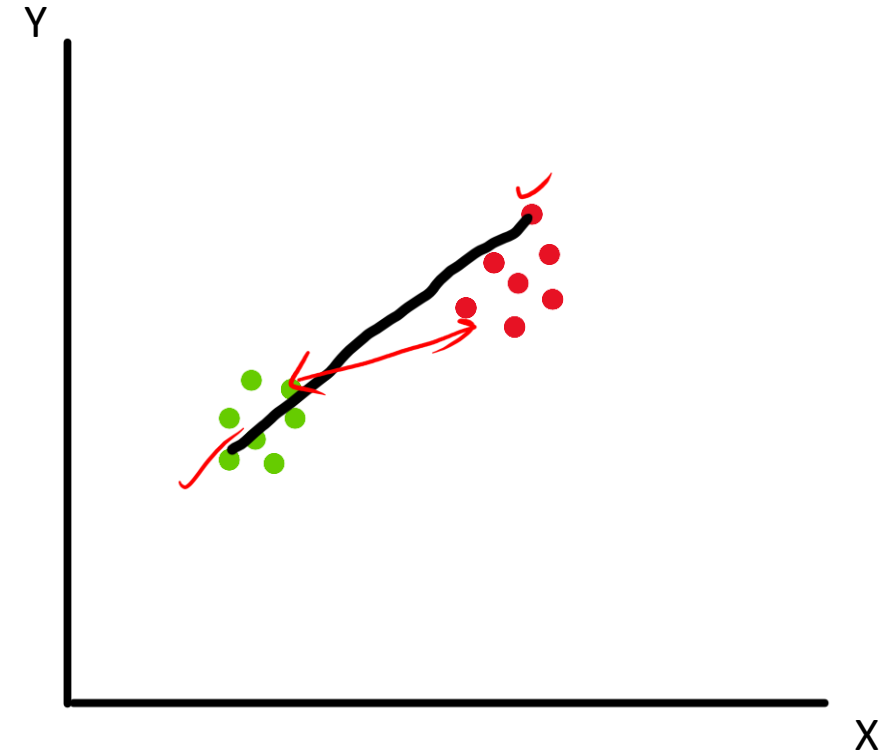
Linkage criteria - Complete Linkage

②

- **Complete Linkage:**

It is the distance between two farthest points of two clusters

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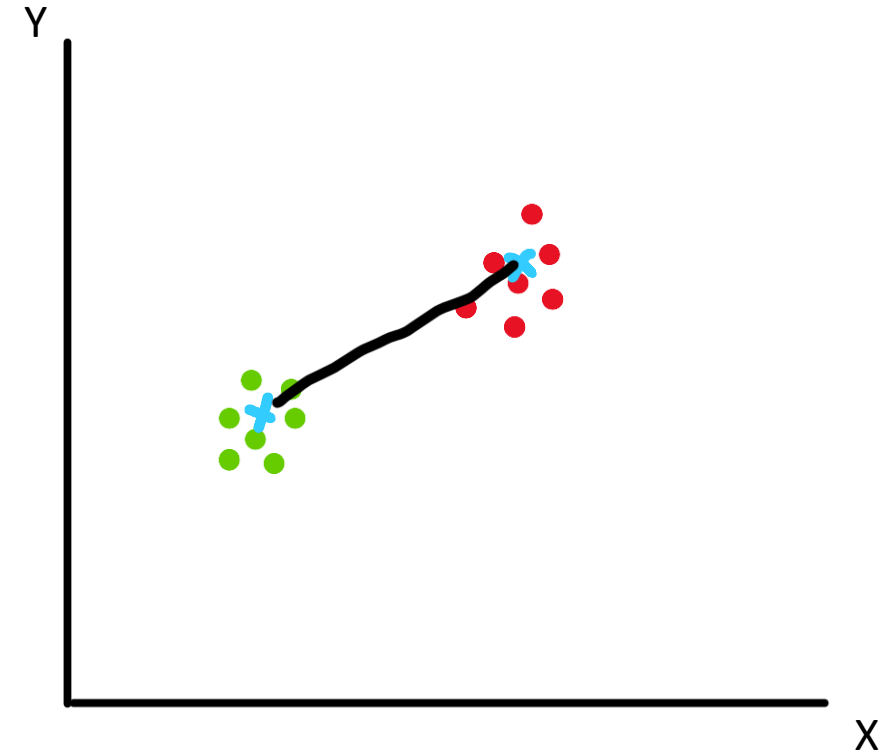


Linkage criteria - Centroid Linkage

③

- **Centroid Linkage:**

It is the distance between centroids of two clusters



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Linkage criteria - Average Linkage

15 distances

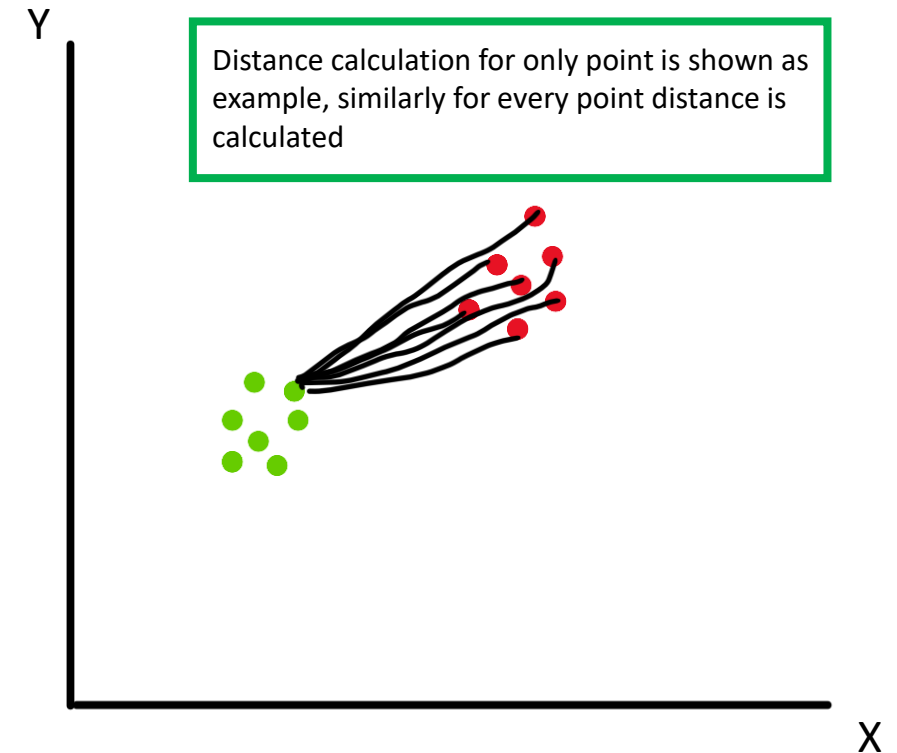
C1 3 points
C2 5 points

Average Linkage:

Distance between each pair of data points in two different clusters is calculated

Total distance is divided by number of data points in two clusters to get average value

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Linkage criteria - Ward Linkage

k-means.
Sum of squared distances

→ calculate centroid
 $d_1, d_2, d_3, \dots$

$$SSE = \sum d_i^2$$

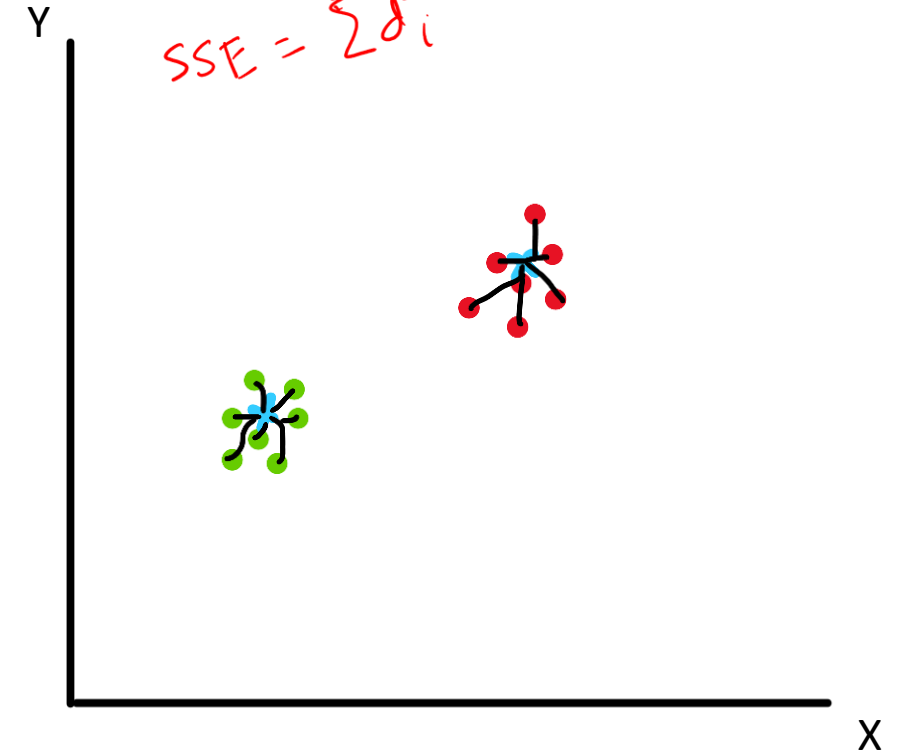
- Ward Linkage:

This linkage criteria tries to minimize sum of squared distance within cluster

Two clusters are merged together / points are assigned to cluster based on how much it will increase the within cluster sum of squared distance from centroid

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This method is least susceptible to noise and outliers





C1

SSD₁



C2

SSD₂



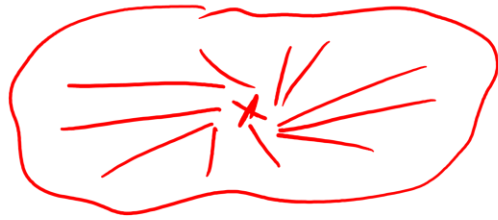
C3

SSD₃

$$\Delta SSD > 0$$

Hypothetically imagine that we merged C1 and C2 $\Rightarrow$ C12

$$\Delta SSD_R = SSD_{12} - (SSD_1 + SSD_2)$$



C12

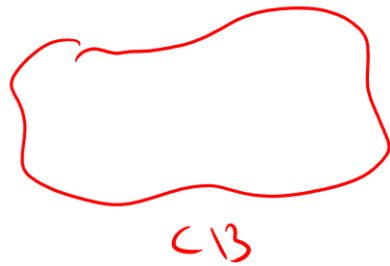
SSD₁₂

ΔSSD_{13}

ΔSSD_{23}

Merge the two clusters together for which ΔSSD is minimum

Let say ΔSSD_{13} is minimum.



Noise and outliers affect the SSD most



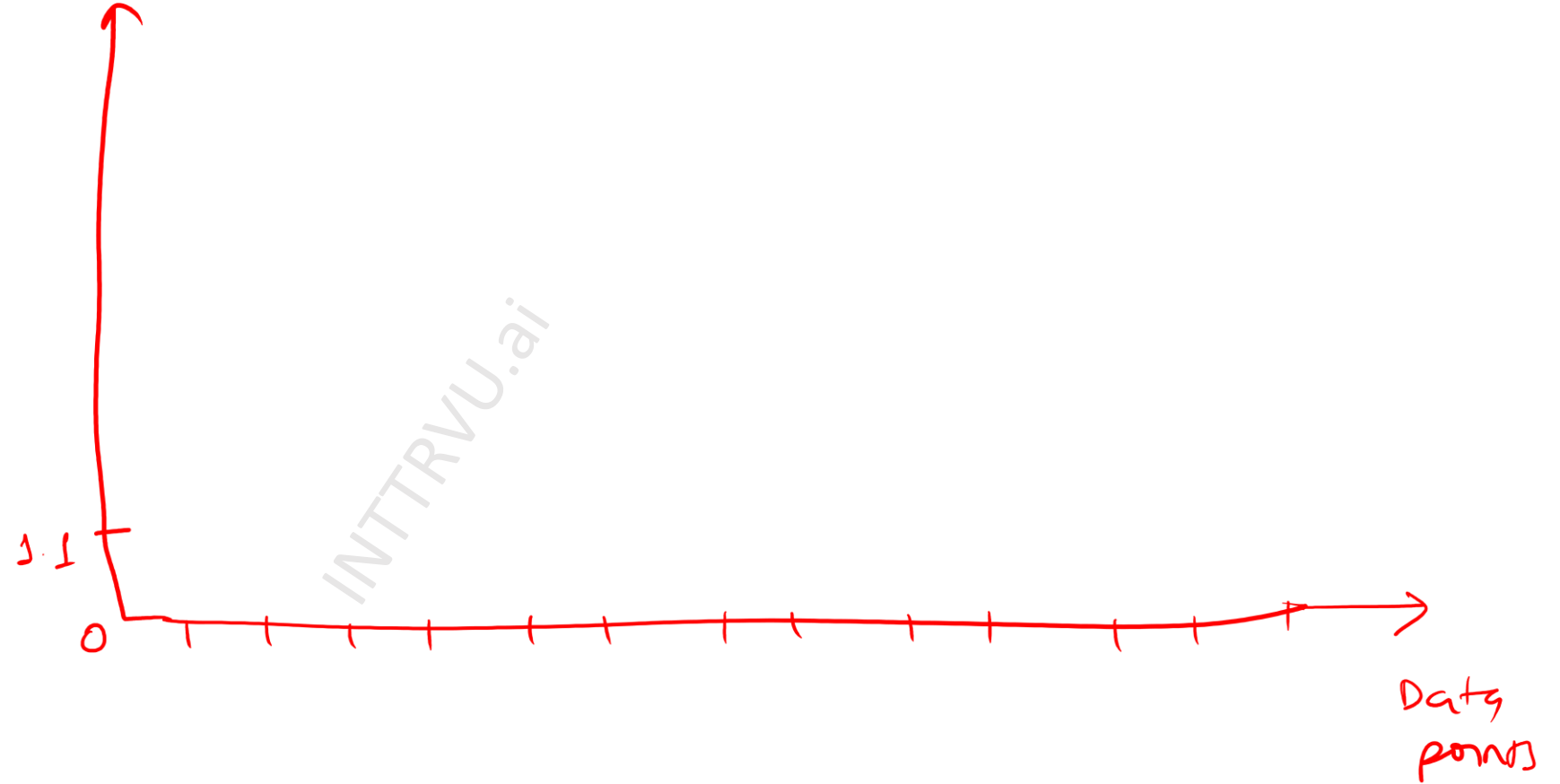
• N

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Distance metric $\rightarrow$ Euclidean
Linkage criteria $\rightarrow$ Ward.

0 $\rightarrow$ 0.1

Distance
Threshold



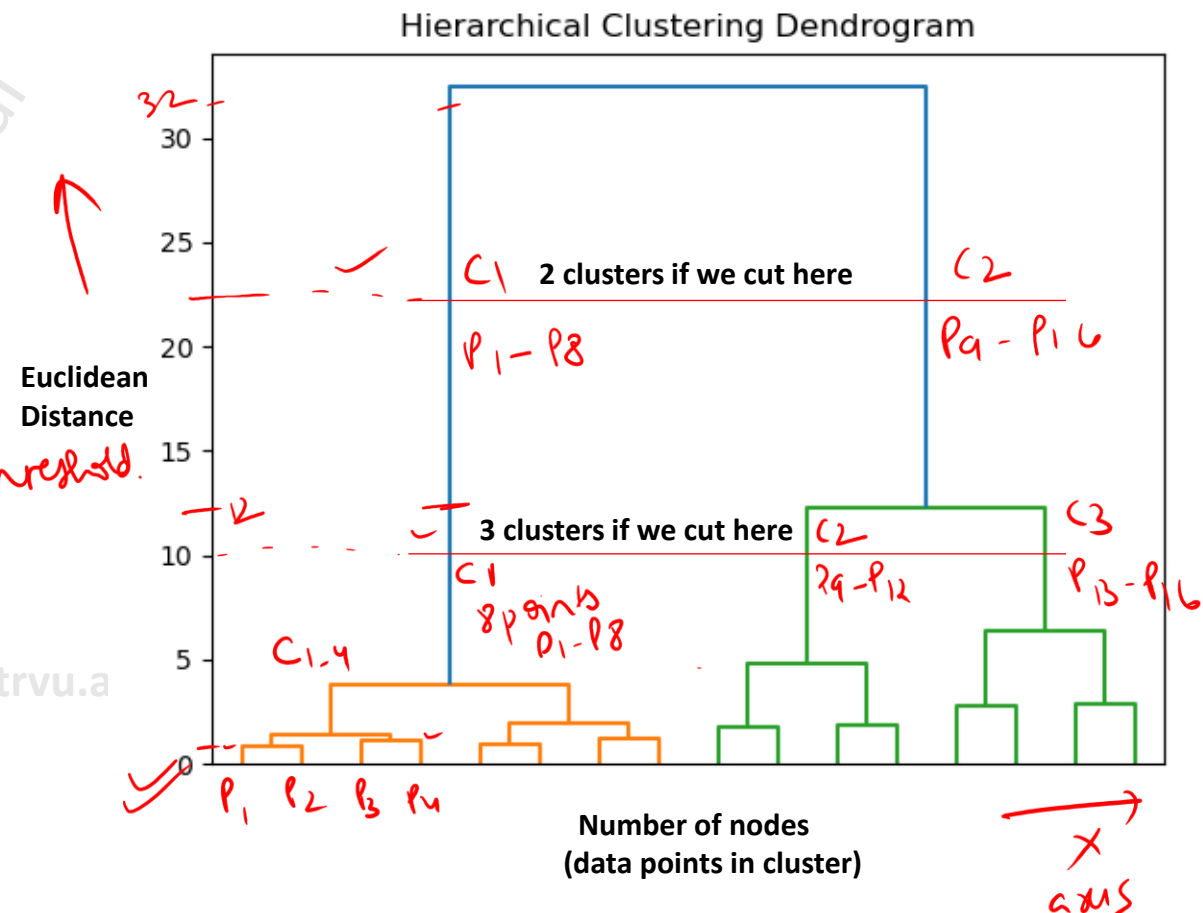
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Hierarchical Clustering - Dendrogram

- Dendrogram is a tree like structure which shows the way in which data is split into groups based on distance
- It is useful in deciding number of clusters
- Dendrogram is cut at appropriate point based on distance threshold to get clusters



Hierarchical Clustering – number of clusters?

Dendrogram is good but for small number of data points.

Option 1:

- Number of clusters can be decided by visualizing dendrogram and cutting at the right point
- But it is very difficult with large real word datasets

① Provide the number of clusters
or
② Provide the distance threshold

Option 2:

- Decide a right linkage criteria and distance metric based on dataset and
- Provide distance threshold parameter (agglomerative clustering implementation in scikit-learn)
- Above this threshold data points/clusters won't be merged

Option 3:

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- Create number of clusters as per business requirement (keep this in mind even if any other method is used)

Advantages and Disadvantages of Hierarchical clustering

Advantages:

- Number of clusters need not be specified ✓
- Options to chose from various distance metrics based on type of data (e.g., cosine distance can be used for text data)
- ✓ Distance threshold suitable for business domain can be chose to get optimized results for the business requirement

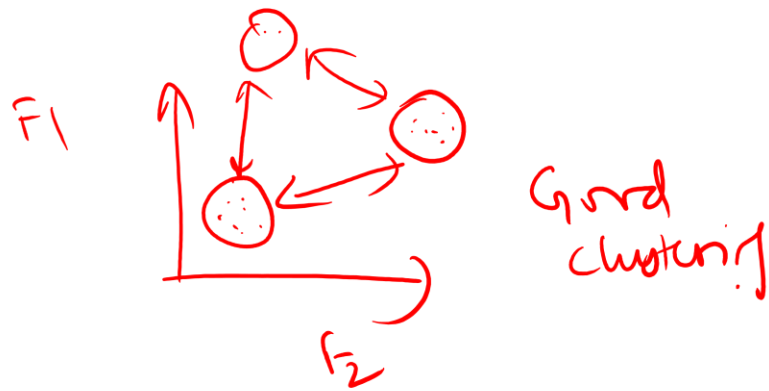
Disadvantages:

- Slow speed of execution for large datasets
- Large memory consumption makes it unsuitable for very large datasets

A good clustering is one where.

① similar data points are grouped together (cohesion)

② dissimilar data points are in different groups. (separation)



Evaluation Methods for Clustering

Silhouette Scoring $\rightarrow$ cohesion and separation

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✓ Silhouette Coefficient is calculated for every data point

Evaluation methods for clustering

✓ Silhouette coefficient

For every data point in the cluster is defined as

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad \checkmark$$

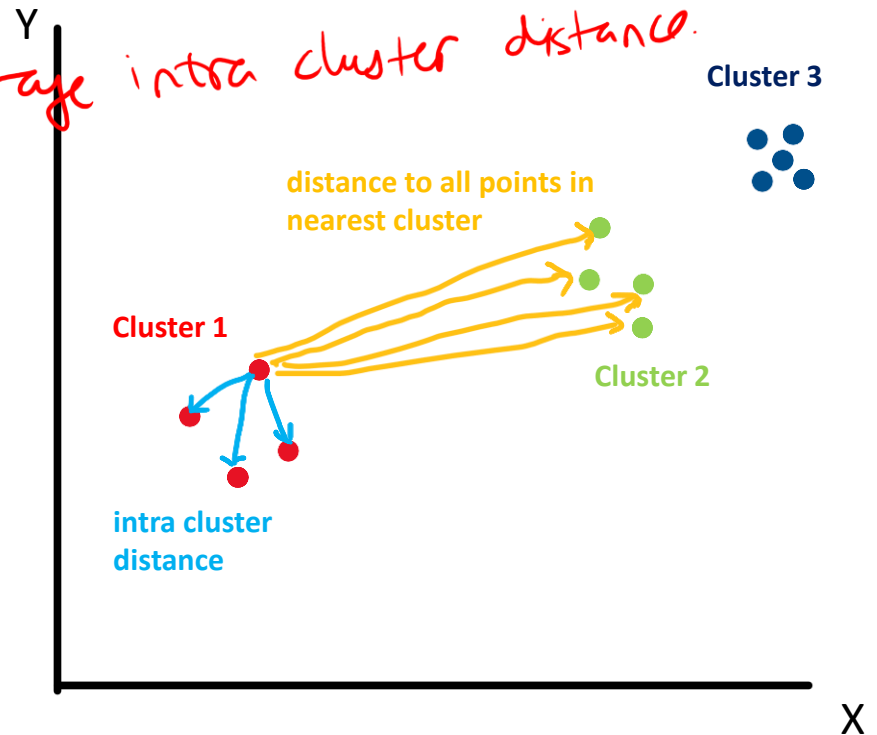
$a(i)$: Average intra-cluster distance

$b(i)$: Average distance to all points in the nearest cluster

- Silhouette coefficient for clustering results is calculated by averaging the above score for all data points
- Value is in the range of -1 to 1
 - 1 indicates worst results (worst results)
 - 0 indicates overlapping of clusters
 - 1 well-formed clusters (best results)

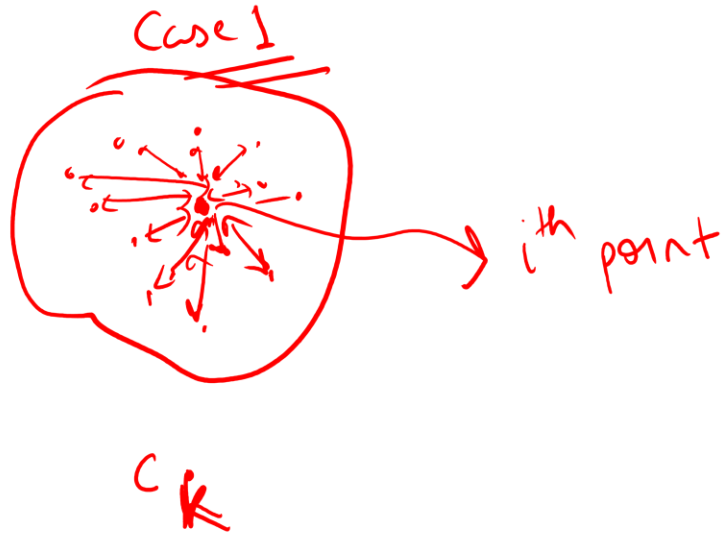
$i \rightarrow i^{\text{th}}$ data point

$a(i) \rightarrow$ Average intra cluster distance.
 $b(i)$



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$a(i)$ ✓ assume i^{th} point belongs to C_k
intra cluster distance (Average)



$$a(i) = \frac{1}{|C_k|} \sum_{\substack{j \in C_k \\ j \neq i}} d_{ji}$$

$|C_k| \rightarrow$ number of data points in cluster k

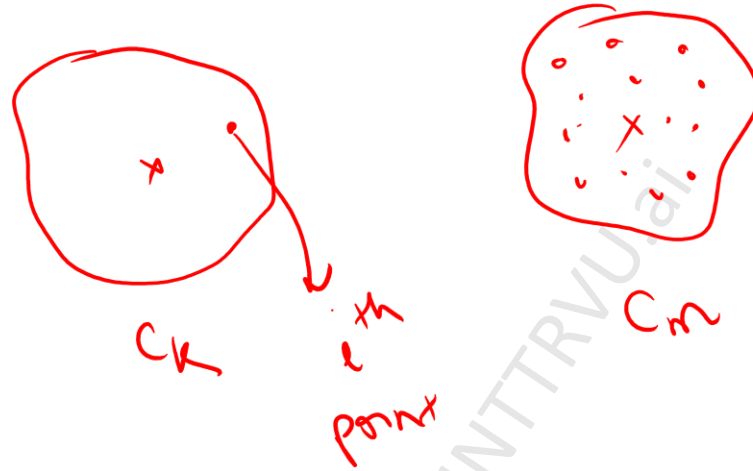


Cohesion

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$b(i)$ $\rightarrow$ separation

Average distance of point i from every point in the nearest cluster

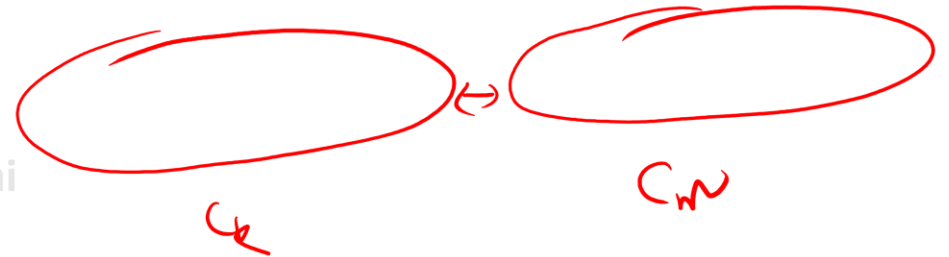


$$b(i) = \frac{1}{|C_m|} \sum_{j \in C_m} d_{ji}$$

Case I



Case 2



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$$S(i) = \frac{b(i) - a(i)}{\max(b(i), a(i))}$$

Case 1 $b(i) > a(i) \Rightarrow S(i) = \frac{b(i) - a(i)}{b(i)} =$

$$1 - \frac{a(i)}{b(i)}$$

very small

Case 2 $b(i) < a(i) \Rightarrow S(i) = \frac{b(i) - a(i)}{a(i)} = \frac{b(i)}{a(i)} - 1$

very small

Ideal condition / cluster $b(i) \gg a(i)$

$\Rightarrow$ Ideal value of $S(i)$ for a good cluster $= 1$ ✓

Bad / worst condition $a(i) \gg b(i)$

worst value of $-S(i) = -1$

$S(i)$ will range from

-1 to 1
 worst value to ideal value

If you have 100 data points

$S(1), S(2), S(3) \dots, S(100)$

Average silhouette coefficient = silhouette score of entire clustering process

$$\langle S \rangle = \frac{1}{100} \sum_{i=1}^{100} S(i)$$

$\langle S \rangle = 1 \rightarrow$ Ideal condition

k range from 3 to 10

$k = 3 \rightarrow \langle S_3 \rangle$

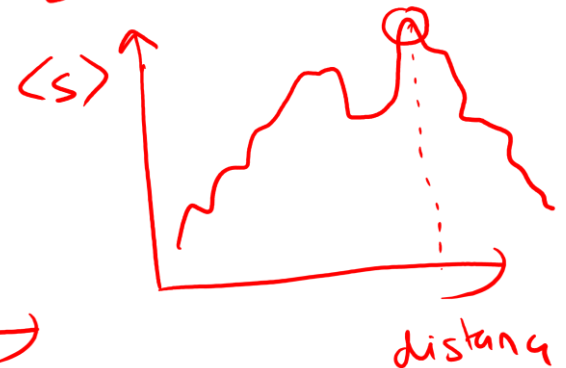
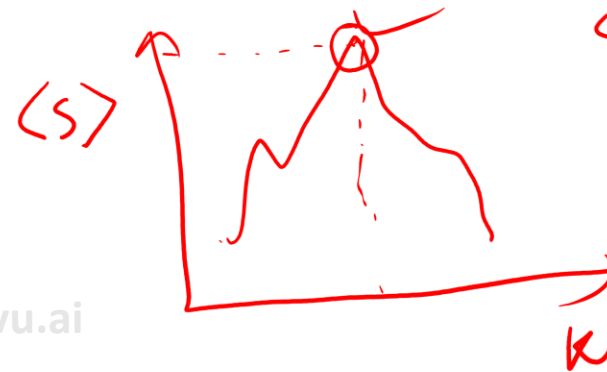
$4 \rightarrow \langle S_4 \rangle$

$5 \rightarrow \langle S_5 \rangle$

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$10 \rightarrow \langle S_{10} \rangle$

distance thresholds.
[0.1 to 10]



Practical Evaluation



Even though Silhouette is a good statistical metric, in practical scenarios it might not be useful

You can use one /both of the following methods to evaluate clustering results

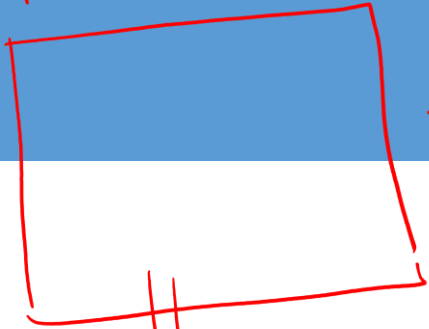
Method 1
Post clustering

- Manual inspection of random samples from large clusters – are they capturing similar pattern?
- Create small labelled data of samples which should be in same cluster and samples which should never be in same cluster and check in which clusters these labelled data points lie after clustering

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Method 2 Pre-clustering.

1M rows



⇒ random sample.

150 rows

⇒ Manual Inspection
Labelled data

clustering

R1	1
R2	2
R3	1
R4	3
R5	4
R100	2

Questions?

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